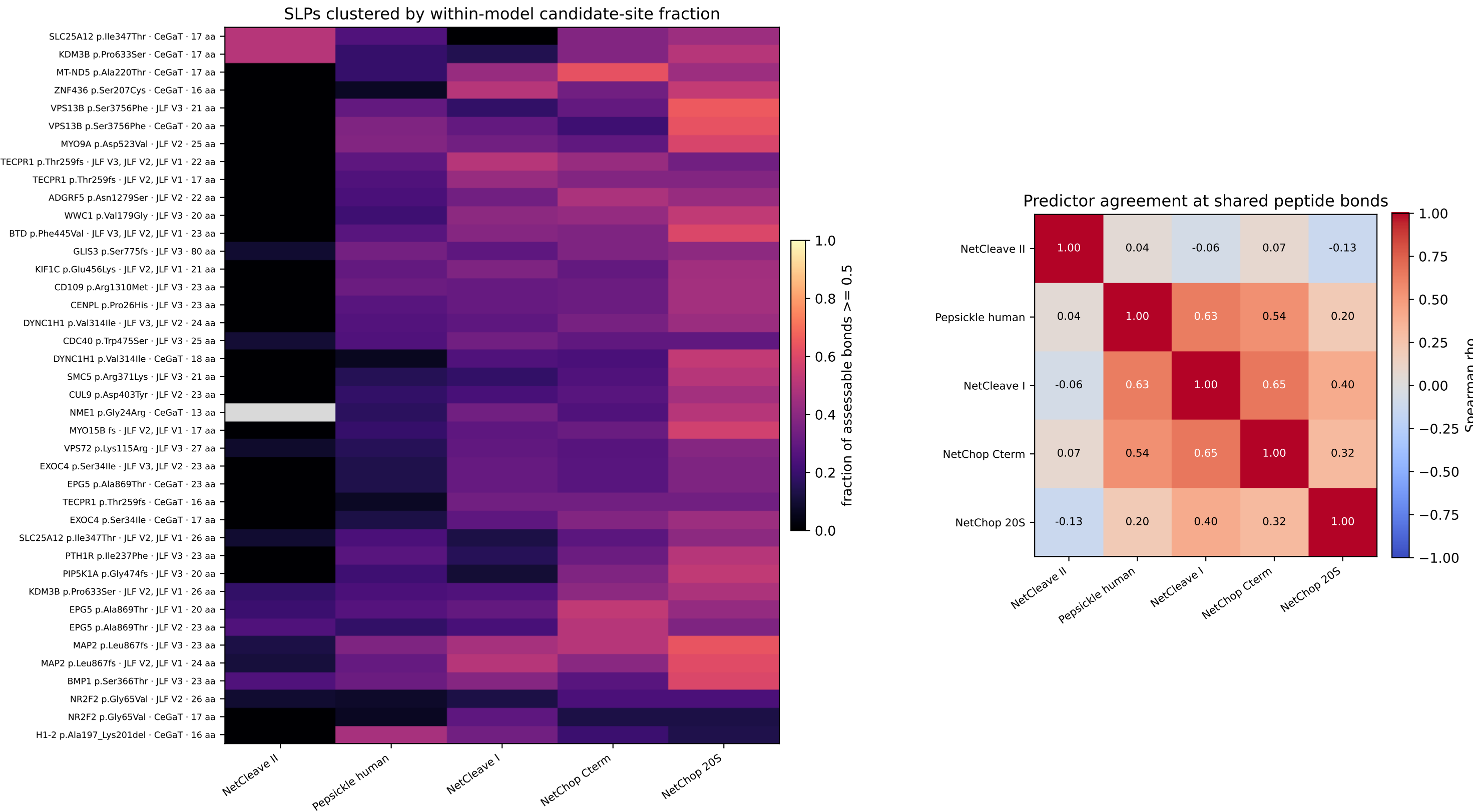


Agreement map for the vaccine SLP cleavage atlas



Clustering organizes the atlas; it does not create an ensemble probability. Sequence pages follow the left-panel SLP order. The figure shows human-only Pepsickle, both biologically distinct NetChop modes, NetCleave-I, and NetCleave-II; the near-redundant all-mammal Pepsickle track remains in the tables. The left panel uses a common display threshold on native 0-1 outputs. The right panel correlates native scores only where both models assess the same bond.

SLC25A12 p.Ile347Thr | CeGaT | 17 aa

disclosed minimal epitope ATAVYPTDL at 3-11

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

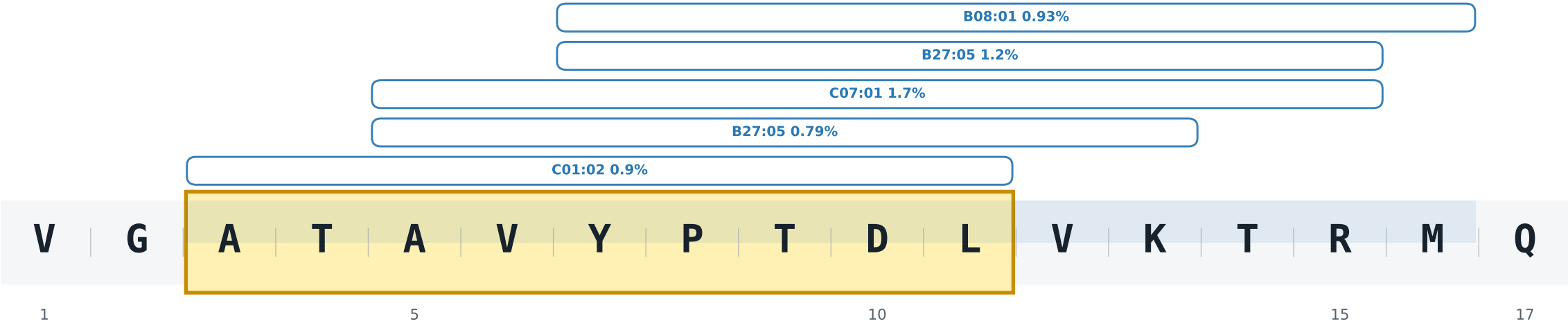
0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

MHC-I: 5 shown / 18 eligible (top-10 cap)

MHC-II: none <= 5% rank



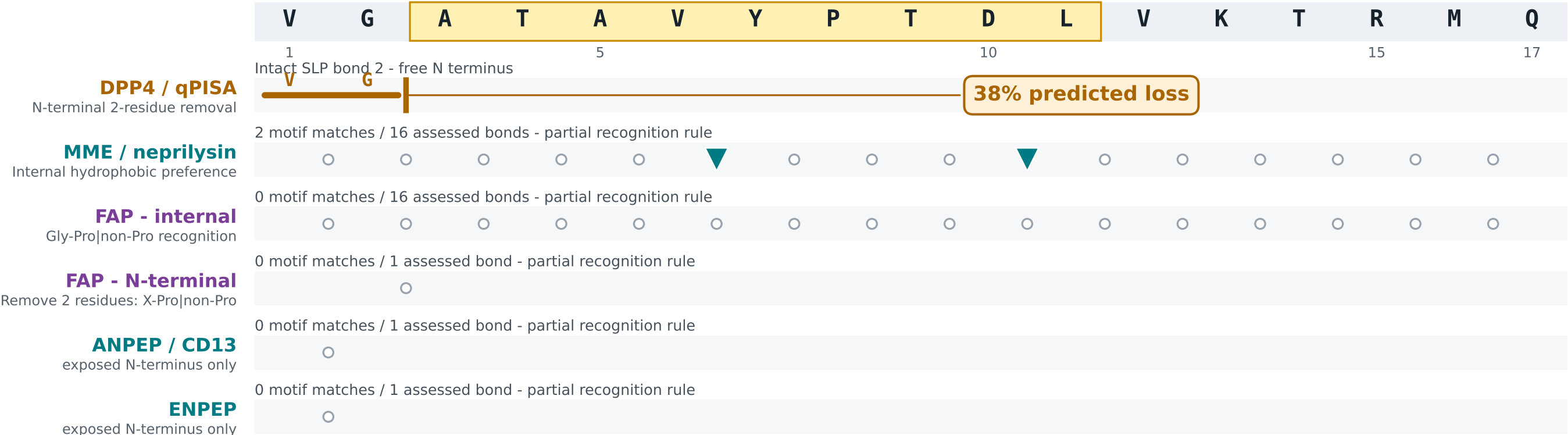
ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

KDM3B p.Pro633Ser | CeGaT | 17 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

NetChop Cterm (ligand) native 0-1

Pepsickle human (in vivo) native 0-1

NetCleave I native 0-1

NetChop 20S (in vitro) native 0-1

0.5 display

0.5 display

0.5 display

0.5 display

MHC-I: 5 shown / 12 eligible (top-10 cap)

C01:02 0.22%

C01:02 1%

B27:05 1.3%

C07:01 0.86%

B27:05 1.2%

P

P

K

L

S

R

E

E

S

S

N

P

F

L

A

F

V

MHC-II: none <= 5% rank

1

5

10

15

17

ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure

P

P

K

L

S

R

E

E

S

S

N

P

F

L

A

F

V

1

5

10

15

17

DPP4 / qPISA

N-terminal 2-residue removal

NOT SCORED / OUTSIDE MODEL SCOPE

Dataset EV2 has missing coefficients for P1=P: P2:P, P1':K

4 motif matches / 16 assessed bonds - partial recognition rule

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MME / neprilysin

Internal hydrophobic preference

0 motif matches / 16 assessed bonds - partial recognition rule

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FAP - internal

Gly-Pro|non-Pro recognition

1 motif match / 1 assessed bond - partial recognition rule

▼

FAP - N-terminal

Remove 2 residues: X-Pro|non-Pro

0 motif matches / 1 assessed bond - partial recognition rule

○

ANPEP / CD13

exposed N-terminus only

0 motif matches / 1 assessed bond - partial recognition rule

○

ENPEP

exposed N-terminus only

Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

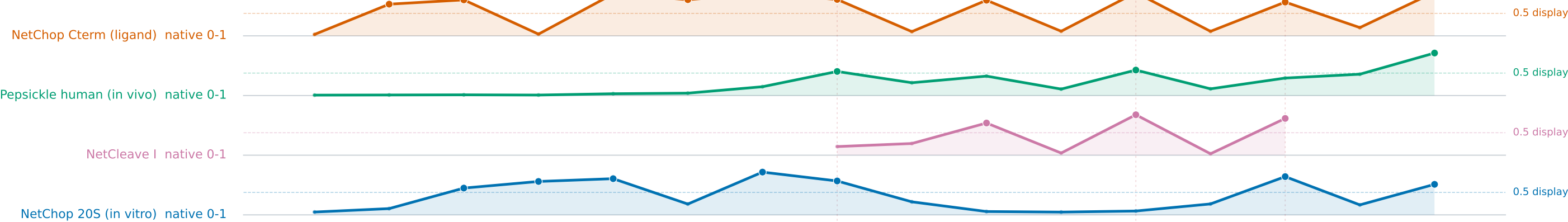
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

MT-ND5 p.Ala220Thr | CeGaT | 17 aa

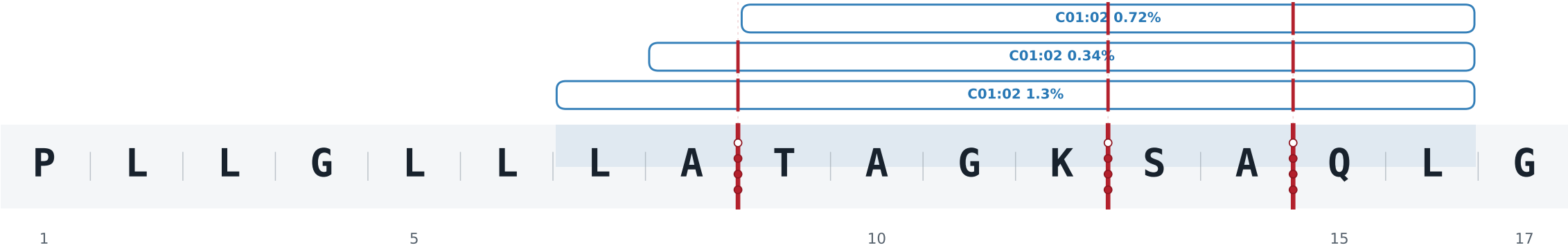
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



MHC-I: 3 shown / 5 eligible (top-10 cap)

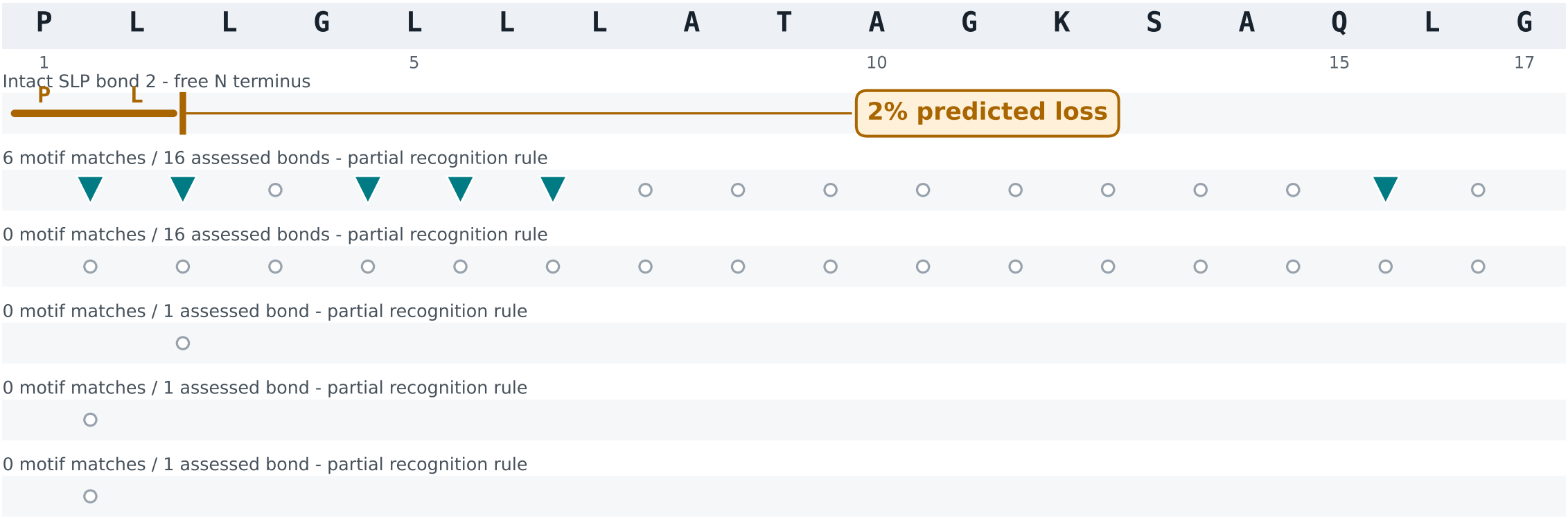


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

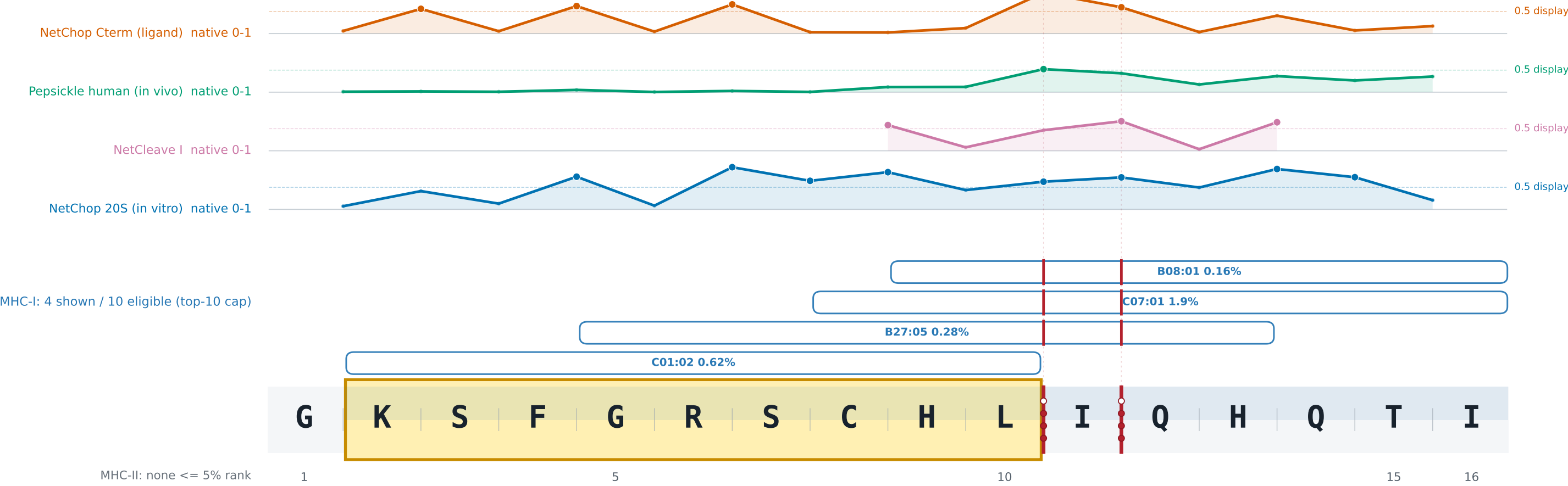
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

ZNF436 p.Ser207Cys | CeGaT | 16 aa

disclosed minimal epitope KSFGRSCHL at 2-10

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

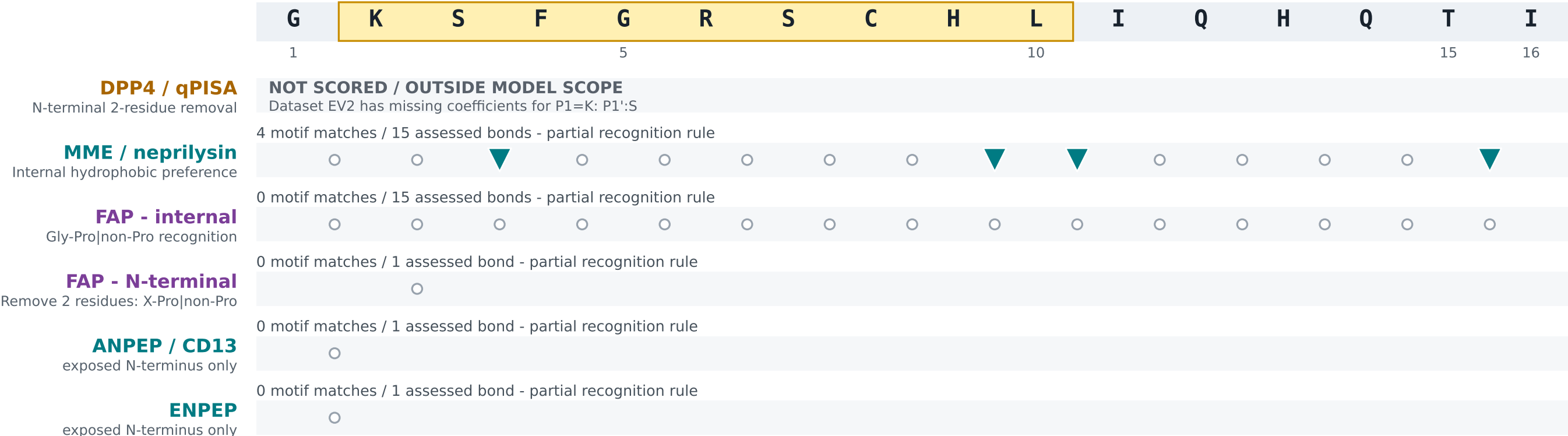


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

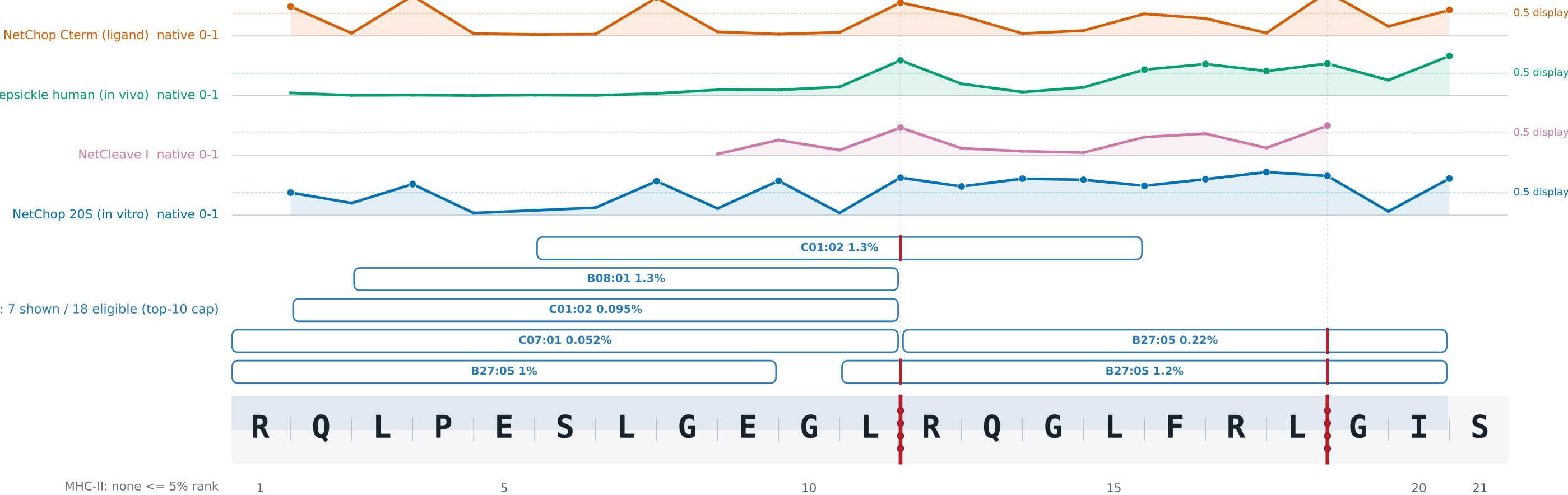
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

VPS13B p.Ser3756Phe| JLF V3 | 21 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

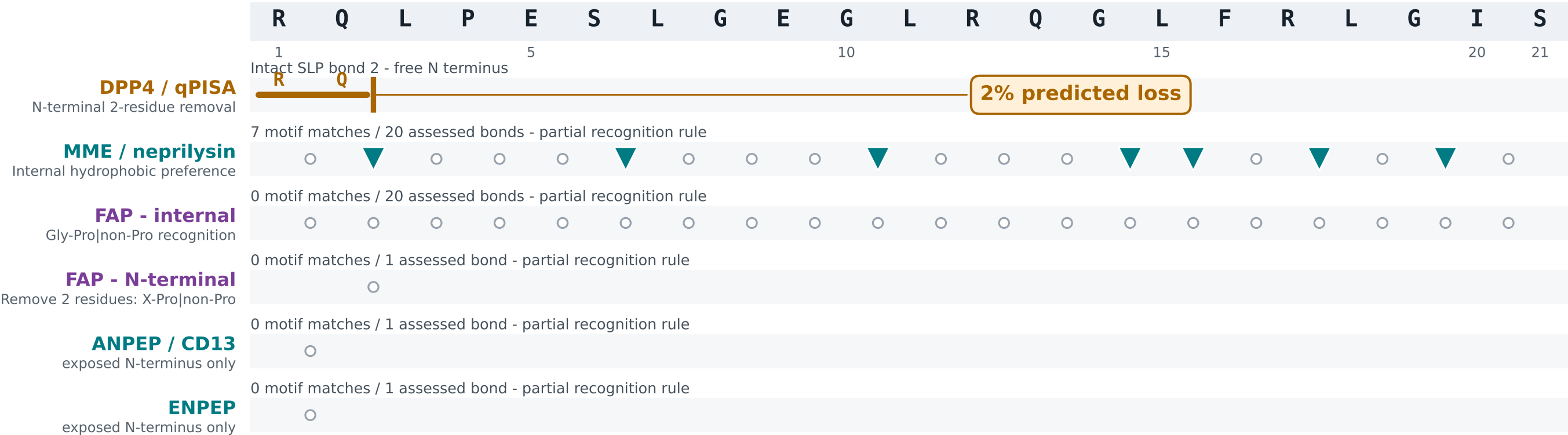


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

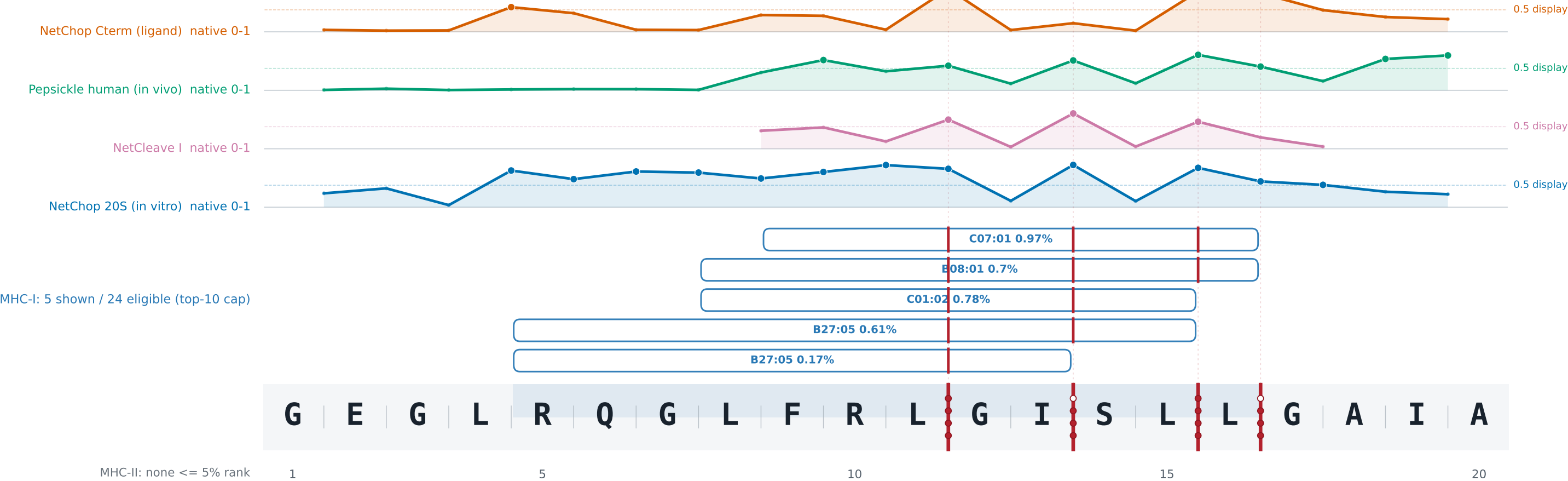
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

VPS13B p.Ser3756Phe | CeGaT | 20 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

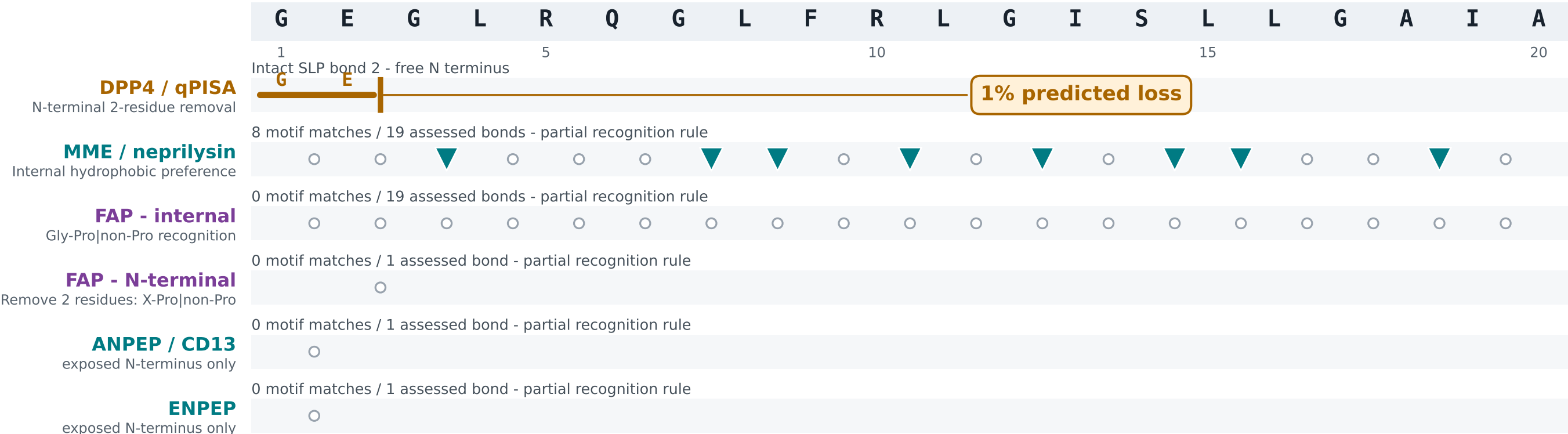


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

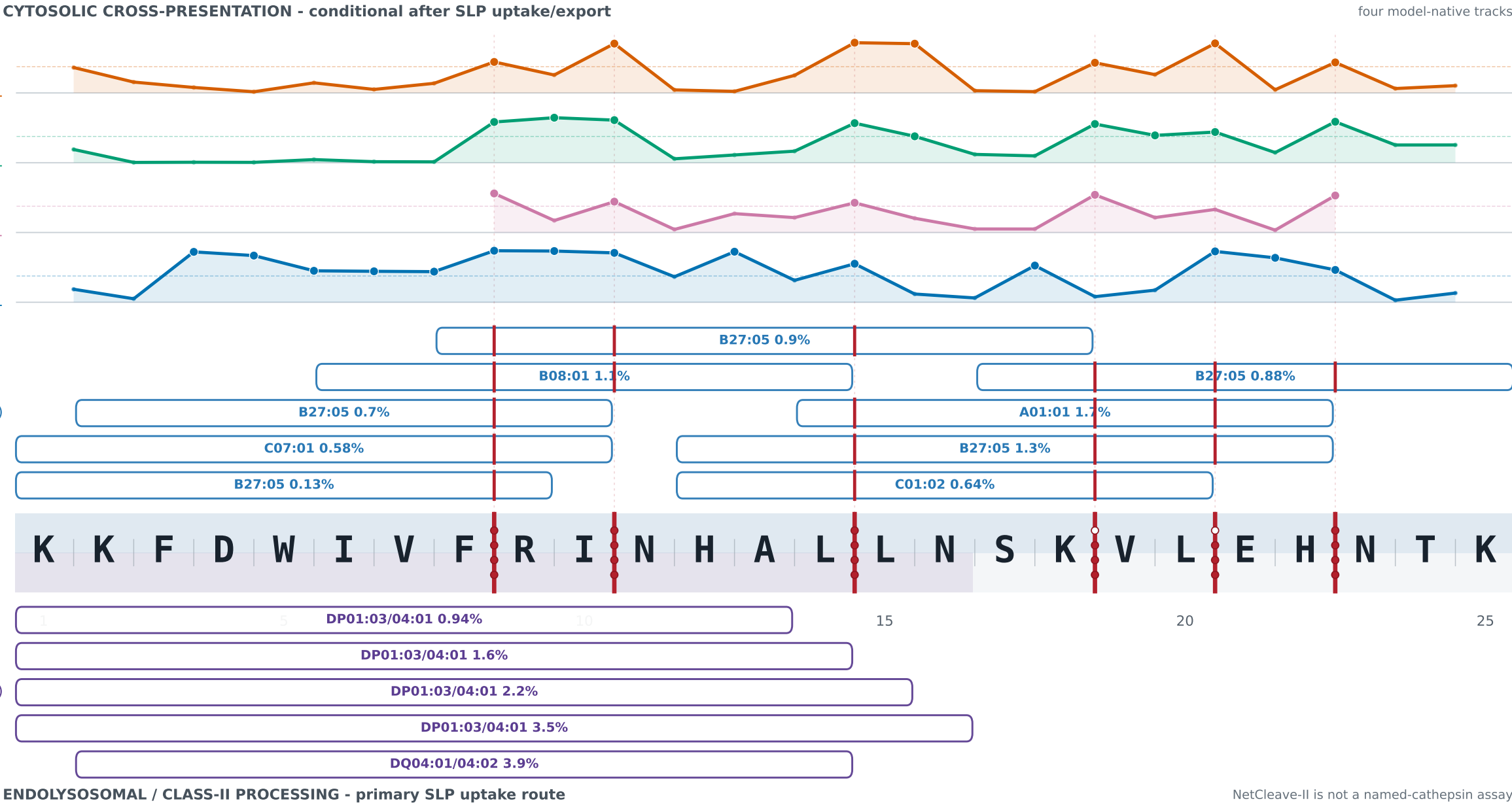
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

MYO9A p.Asp523Val | JLF V2 | 25 aa

minimal epitope/window not disclosed for this SLI



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

TECPR1 p.Thr259fs | JLF V3, JLF V2, JLF V1 | 22 aa

disclosed minimal epitope TIPAAPVGL at 9-17

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

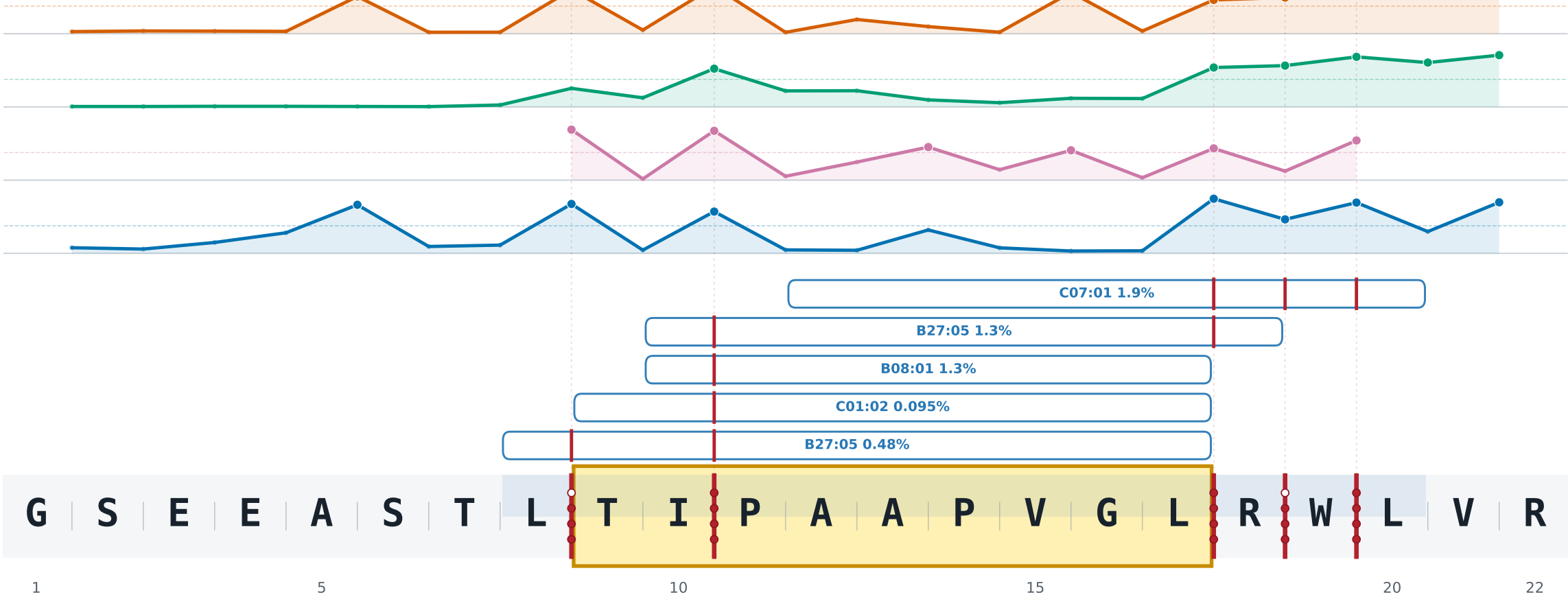
0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

MHC-I: 5 shown / 15 eligible (top-10 cap)

MHC-II: none <= 5% rank



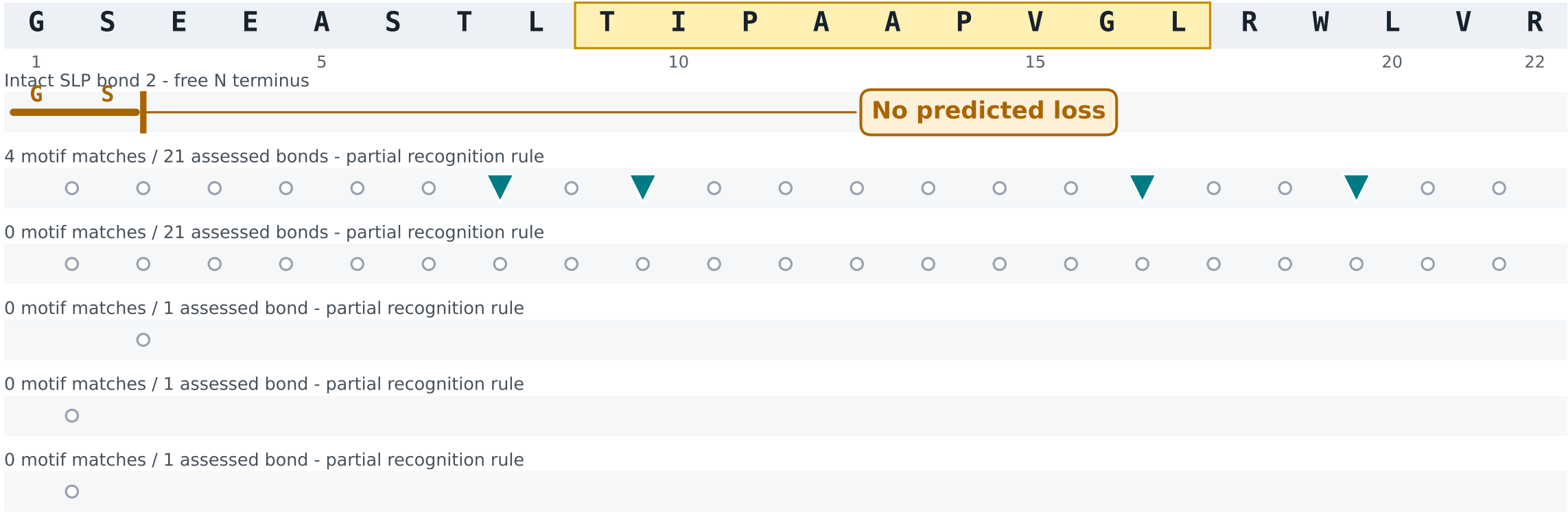
ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

TECPR1 p.Thr259fs | JLF V2, JLF V1 | 17 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

NetChop Cterm (ligand) native 0-1

Pepsickle human (in vivo) native 0-1

NetCleave I native 0-1

NetChop 20S (in vitro) native 0-1

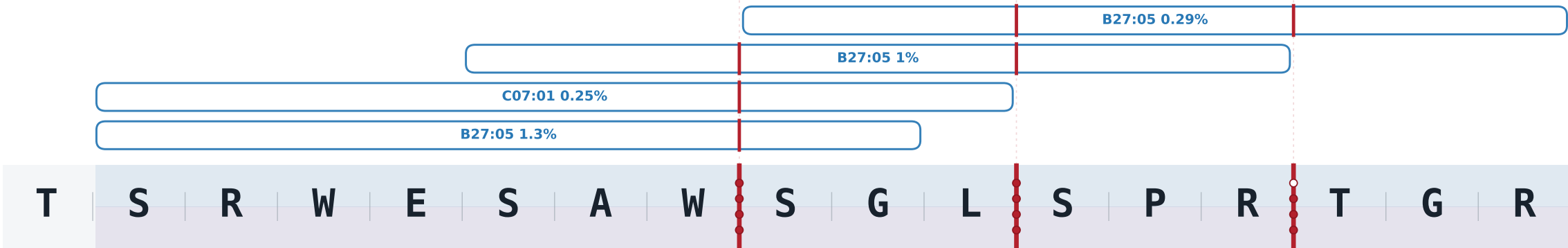
0.5 display

0.5 display

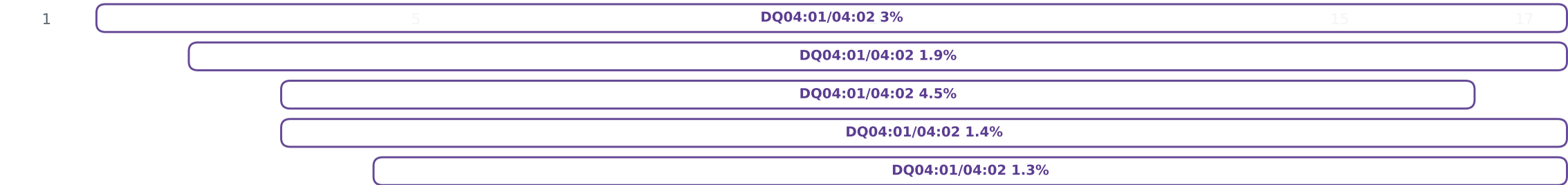
0.5 display

0.5 display

MHC-I: 4 shown / 5 eligible (top-10 cap)



MHC-II: 5 shown / 5 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure

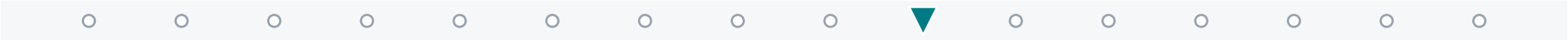
T	S	R	W	E	S	A	W	S	G	L	S	P	R	T	G	R
1				5					10					15		17

DPP4 / qPISA
N-terminal 2-residue removal

NOT SCORED / OUTSIDE MODEL SCOPE
Dataset EV2 has missing coefficients for P1=S: P1':R

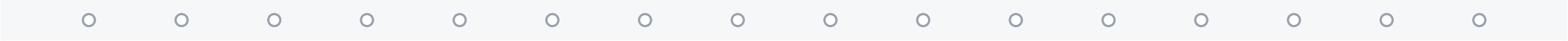
MME / neprilysin
Internal hydrophobic preference

1 motif match / 16 assessed bonds - partial recognition rule



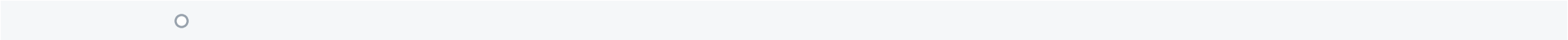
FAP - internal
Gly-Pro|non-Pro recognition

0 motif matches / 16 assessed bonds - partial recognition rule



FAP - N-terminal
Remove 2 residues: X-Pro|non-Pro

0 motif matches / 1 assessed bond - partial recognition rule



ANPEP / CD13
exposed N-terminus only

0 motif matches / 1 assessed bond - partial recognition rule



ENPEP
exposed N-terminus only

0 motif matches / 1 assessed bond - partial recognition rule



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

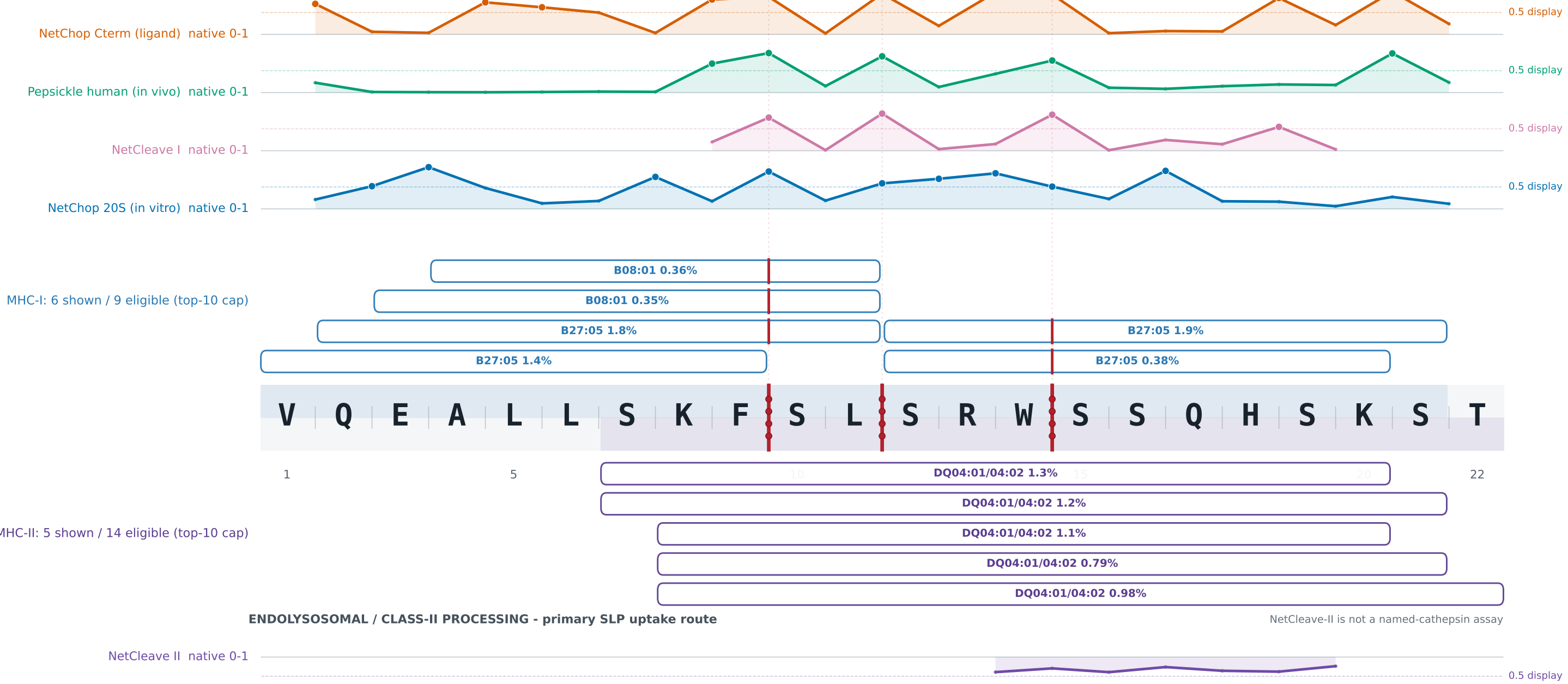
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

ADGRF5 p.Asn1279Ser | JLF V2 | 22 aa

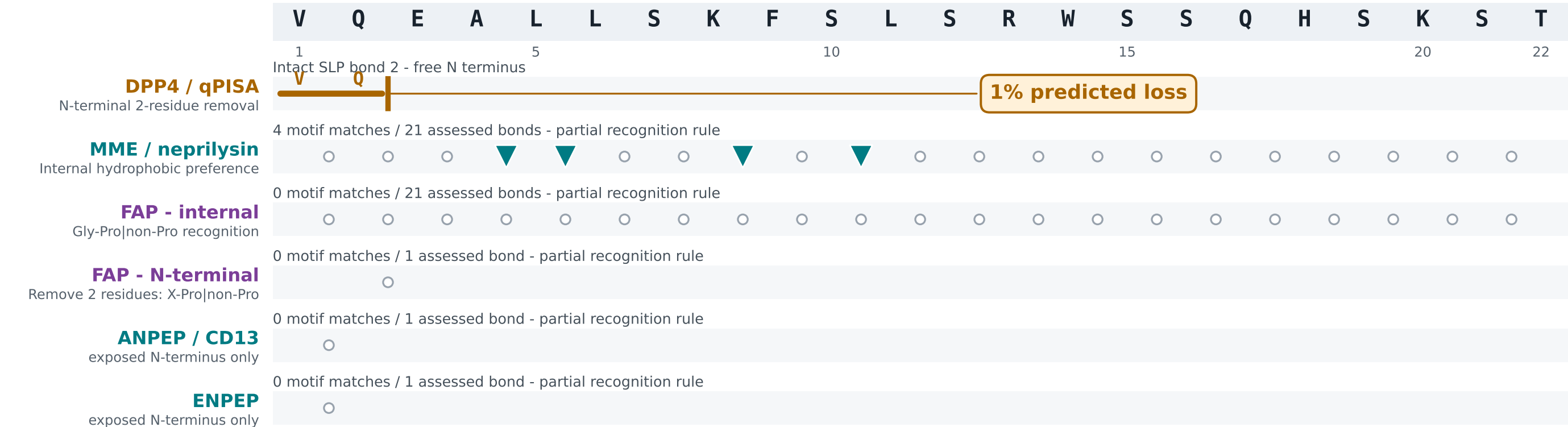
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

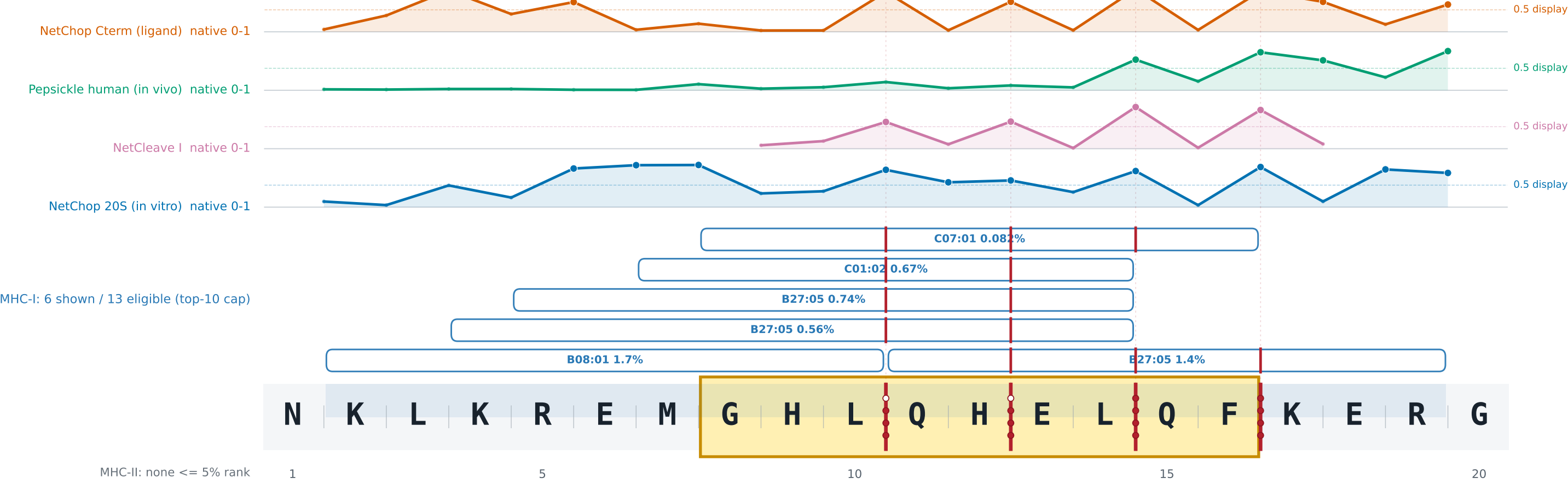
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

WWC1 p.Val179Gly | JLF V3 | 20 aa

disclosed minimal epitope GHLQHELQF at 8-16

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

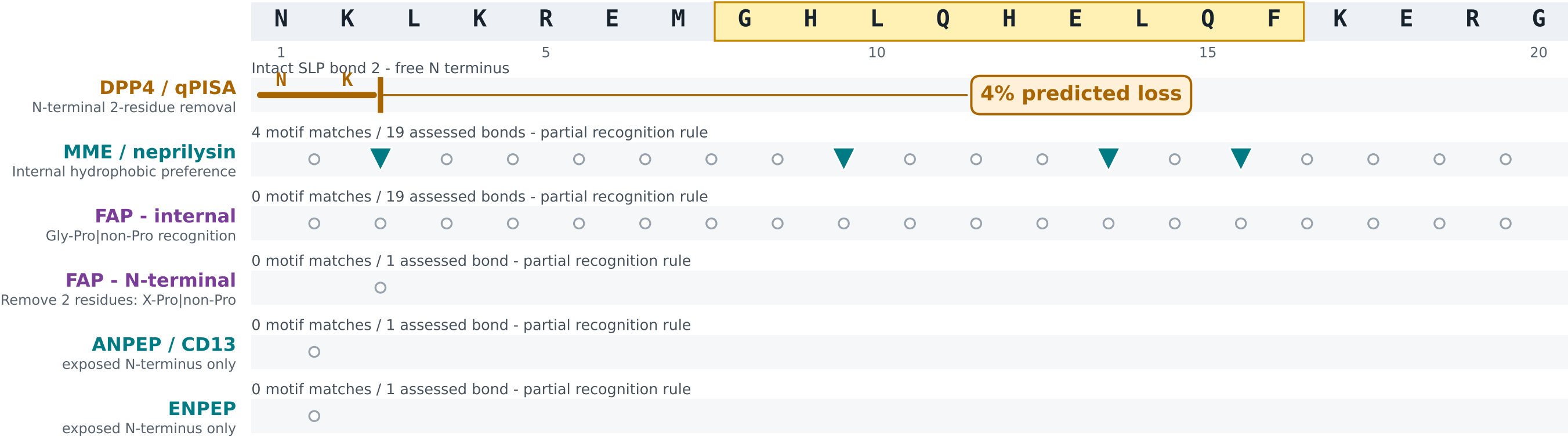


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

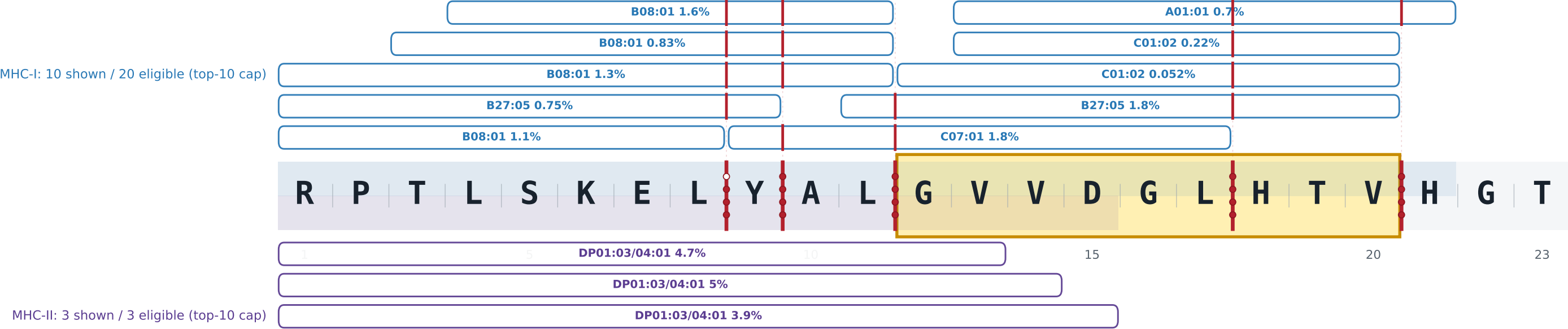
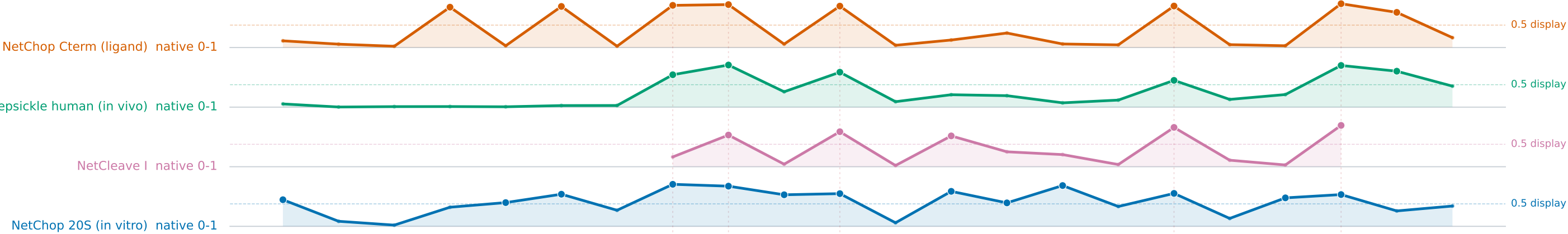
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

BTD p.Phe445Val | JLF V3, JLF V2, JLF V1 | 23 aa

disclosed minimal epitope GVVDGLHTV at 12-20

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

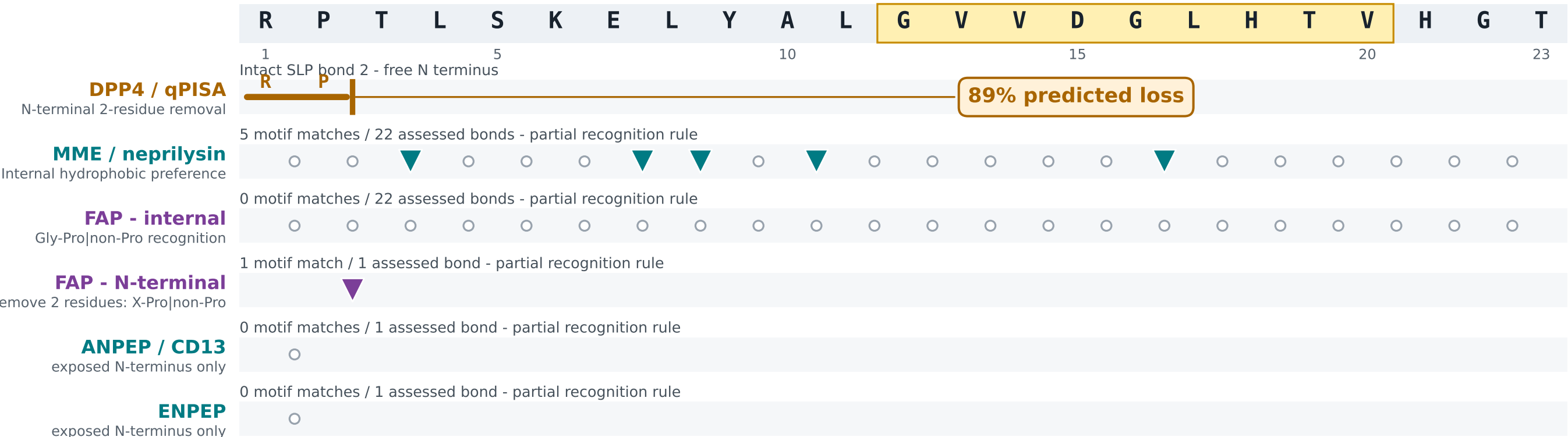


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

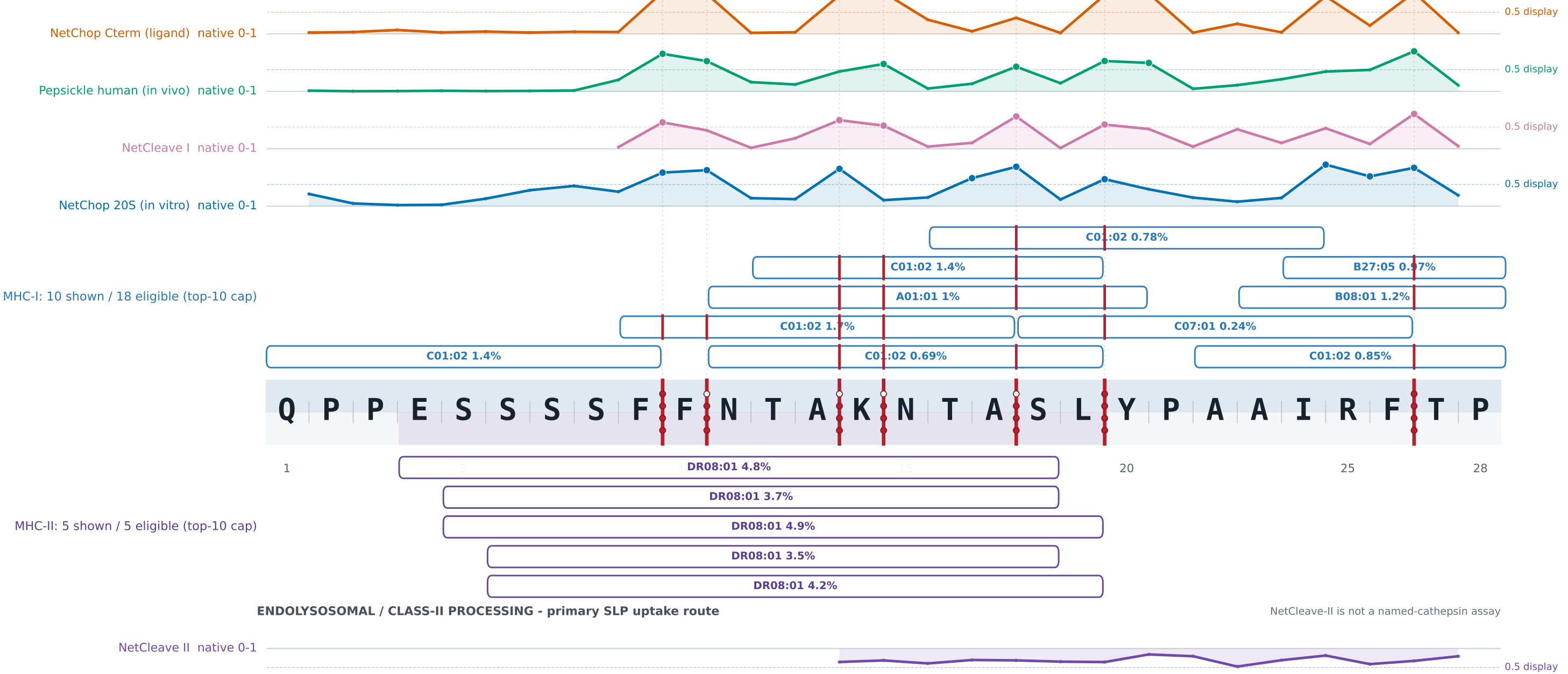
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

GLIS3 p.Ser775fs | JLF V3 | 80 aa - segment 1/3, bonds 1-27

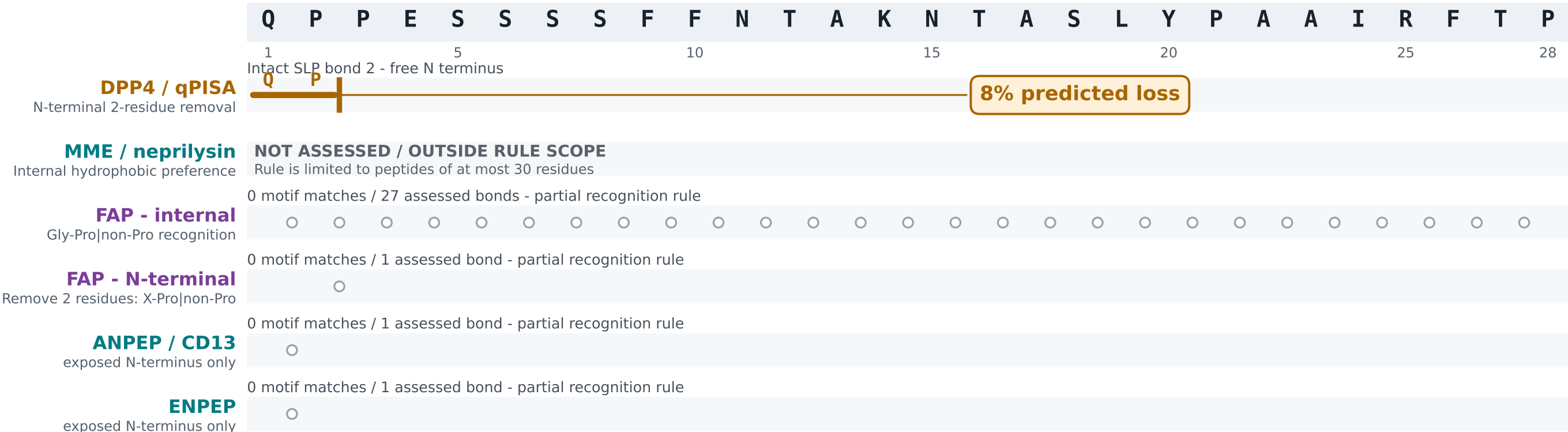
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

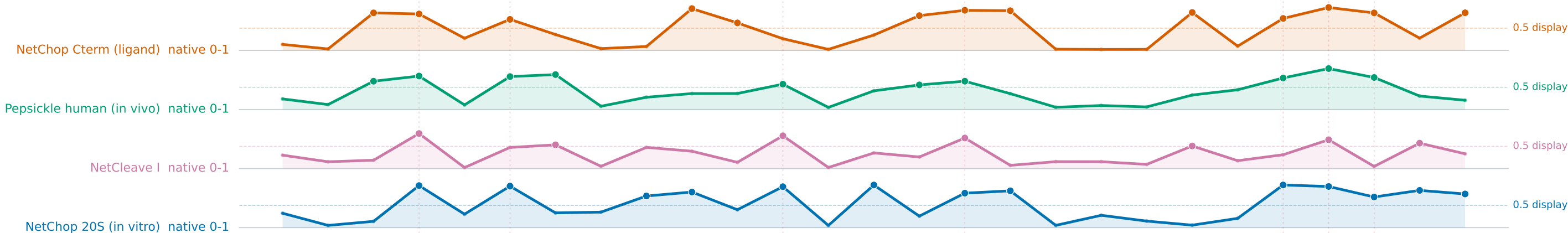
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

GLIS3 p.Ser775fs | JLF V3 | 80 aa - segment 2/3, bonds 28-54

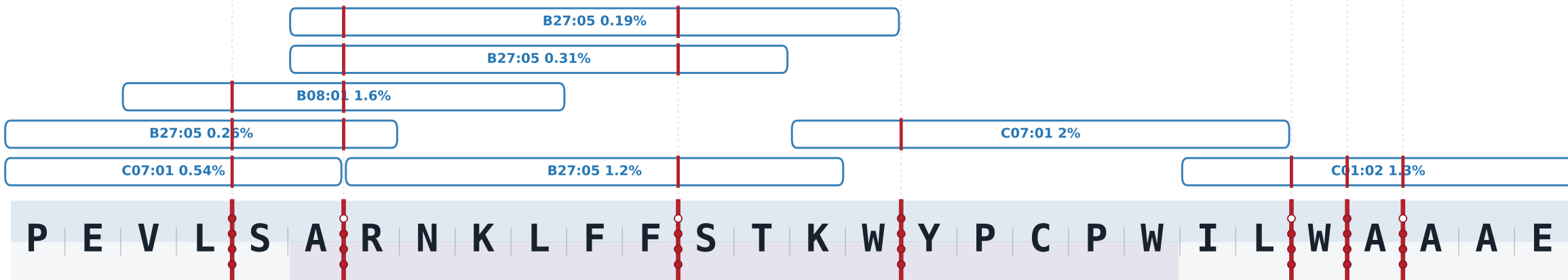
minimal epitope/window not disclosed for this SLI

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

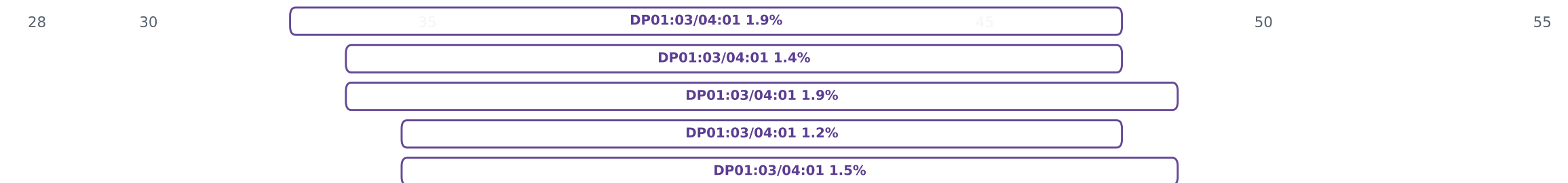
four model-native tracks



MHC-I: 8 shown / 14 eligible (top-10 cap)

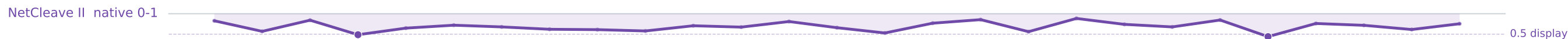


MHC-II: 5 shown / 19 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-Il is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



DPP4 / qPISA

N-terminal 2-residue removal

SITE OUTSIDE THIS SEGMENT | full SLP: 8% predicted loss

Intact SLP bond 2 is outside this segment

MME / neprilysin

Internal hydrophobic preference

NOT ASSESSED / OUTSIDE RULE SCOPE

Rule is limited to peptides of at most 30 residues

0 motif matches / 27 assessed bonds - partial recognition rule

FAP - interna

Gly-Pro|non-Pro recognition



FAP - N-termina

Remove 2 residues: X-Pro|non-Pro

SITE OUTSIDE THIS SEGMENT

Intact N-terminal bond 2 is outside this segment

ANPEP / CD13

exposed N-terminus only

SITE OUTSIDE THIS SEGMENT

Intact N-terminal bond 1 is outside this segment

ENPEF

exposed N-terminus only

SITE OUTSIDE THIS SEGMENT

Intact N-terminal bond 1 is outside this segment

Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only

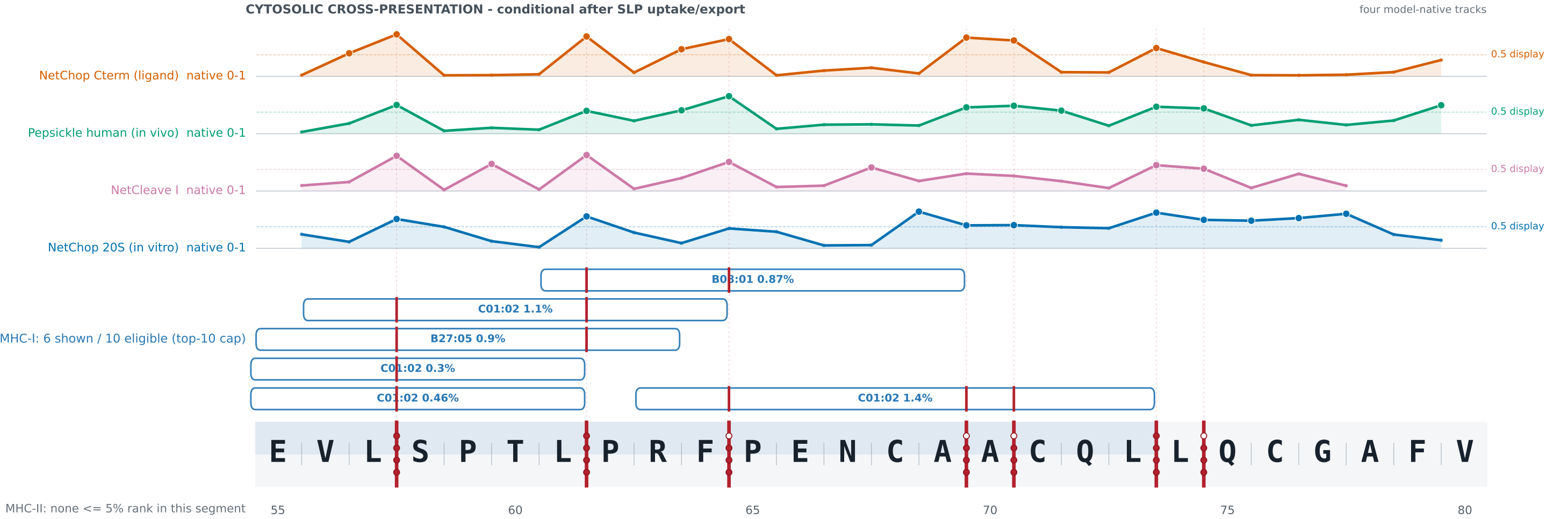
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV

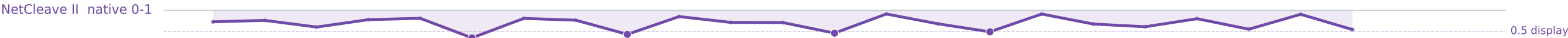
GLIS3 p.Ser775fs | JLF V3 | 80 aa - segment 3/3, bonds 55-79

minimal epitope/window not disclosed for this SLP



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



DPP4 / qPISA N-terminal 2-residue removal	SITE OUTSIDE THIS SEGMENT full SLP: 8% predicted loss Intact SLP bond 2 is outside this segment
MME / neprilysin Internal hydrophobic preference	NOT ASSESSED / OUTSIDE RULE SCOPE Rule is limited to peptides of at most 30 residues
FAP - internal Gly-Pro non-Pro recognition	0 motif matches / 25 assessed bonds - partial recognition rule
FAP - N-terminal Remove 2 residues: X-Pro non-Pro	SITE OUTSIDE THIS SEGMENT Intact N-terminal bond 2 is outside this segment
ANPEP / CD13 exposed N-terminus only	SITE OUTSIDE THIS SEGMENT Intact N-terminal bond 1 is outside this segment
ENPEP exposed N-terminus only	SITE OUTSIDE THIS SEGMENT Intact N-terminal bond 1 is outside this segment

Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

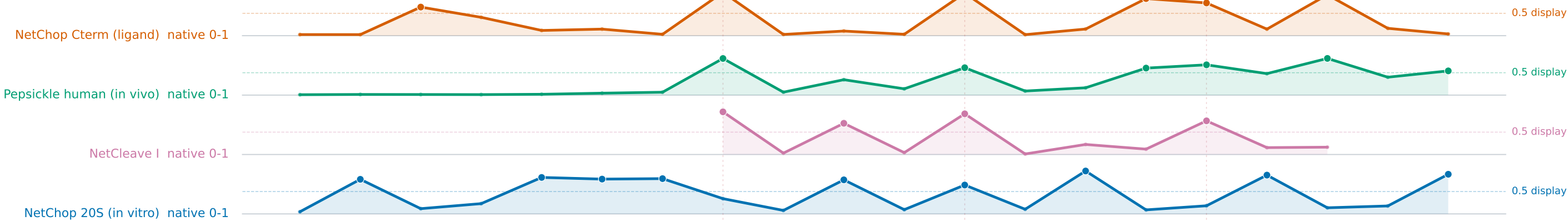
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

KIF1C p.Glu456Lys | JLF V2, JLF V1 | 21 aa

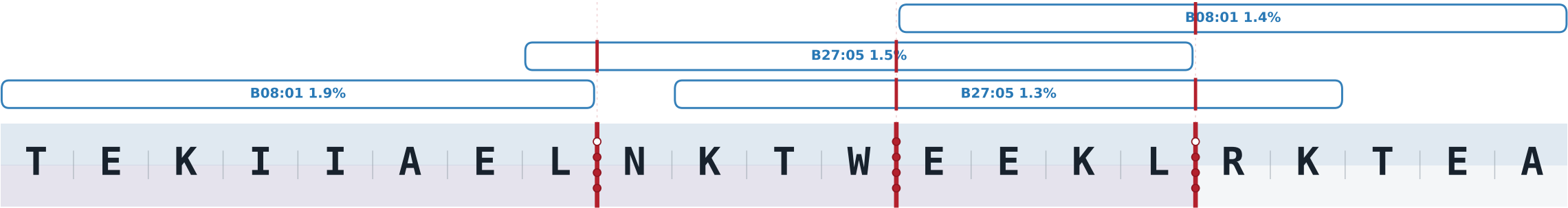
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

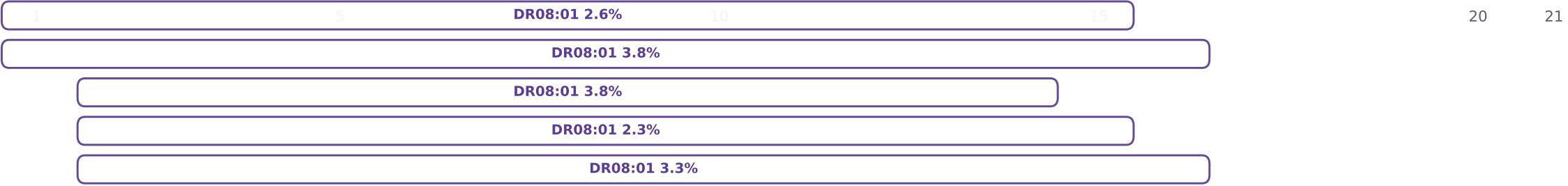


MHC-I: 4 shown / 4 eligible (top-10 cap)



T E K I I A E L N K T W E E K L R K T E A

MHC-II: 5 shown / 8 eligible (top-10 cap)



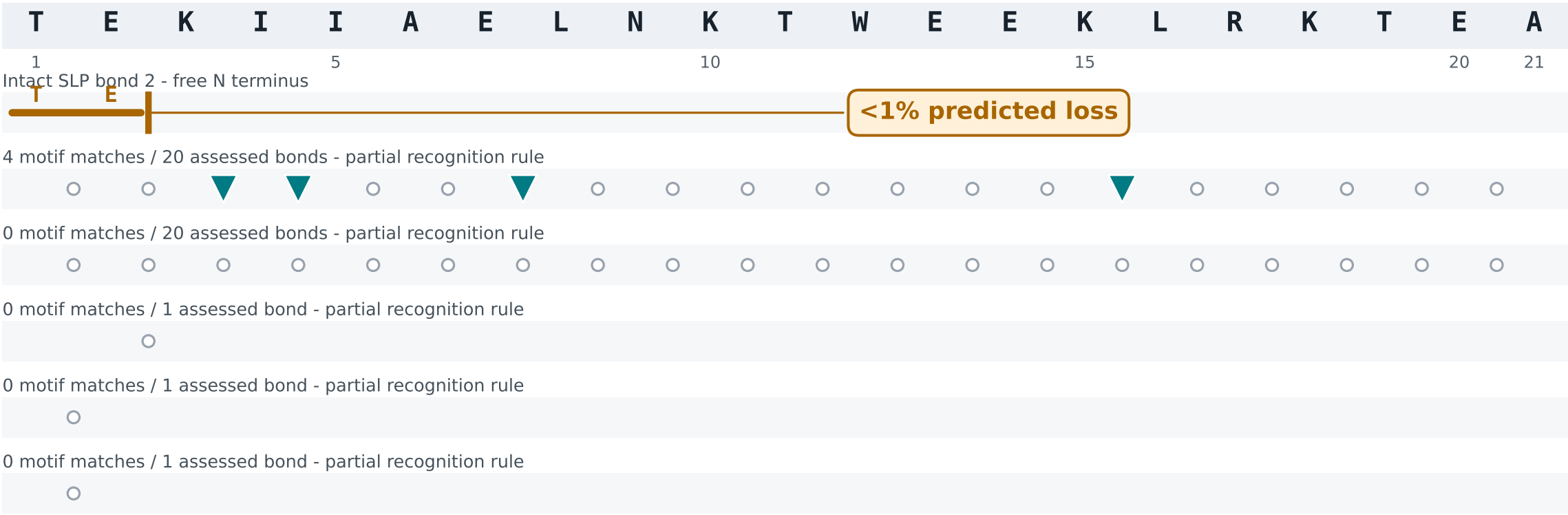
ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

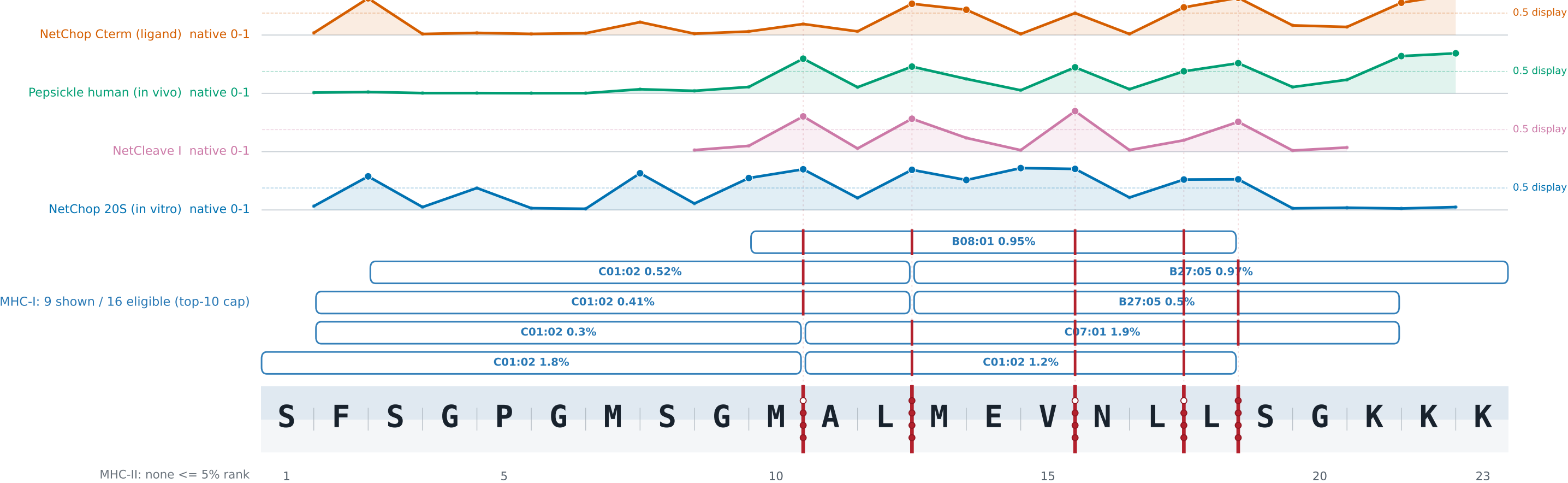
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

CD109 p.Arg1310Met | JLF V3 | 23 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



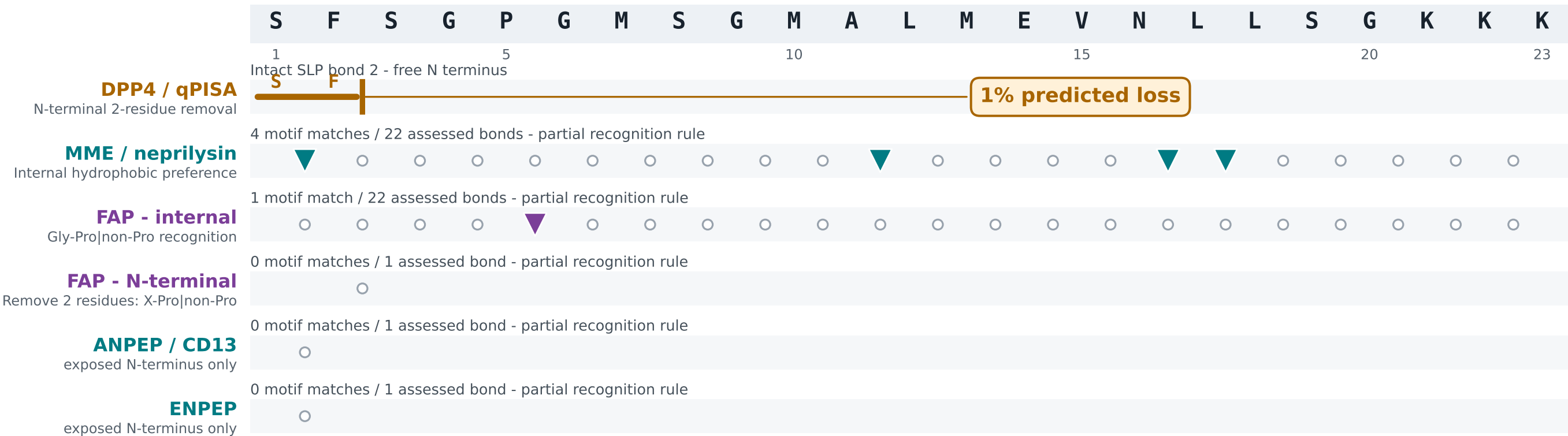
ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

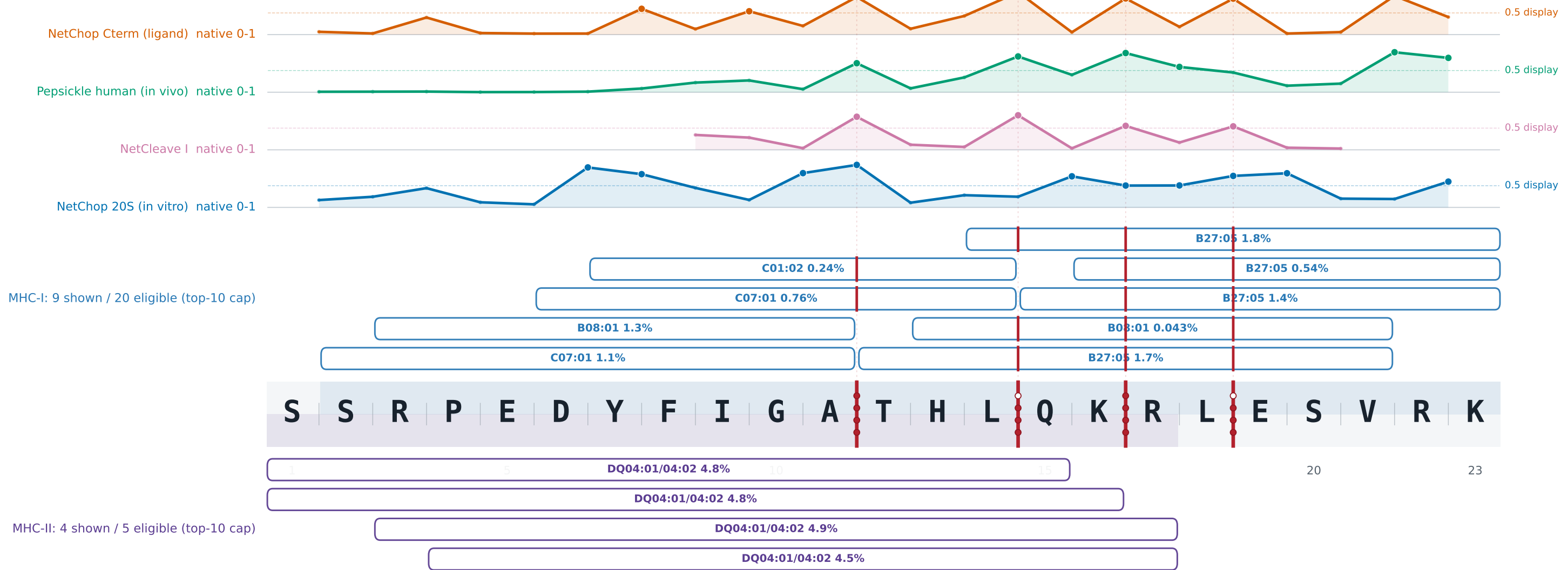
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

CENPL p.Pro26His | JLF V3 | 23 aa

minimal epitope/window not disclosed for this SLI

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

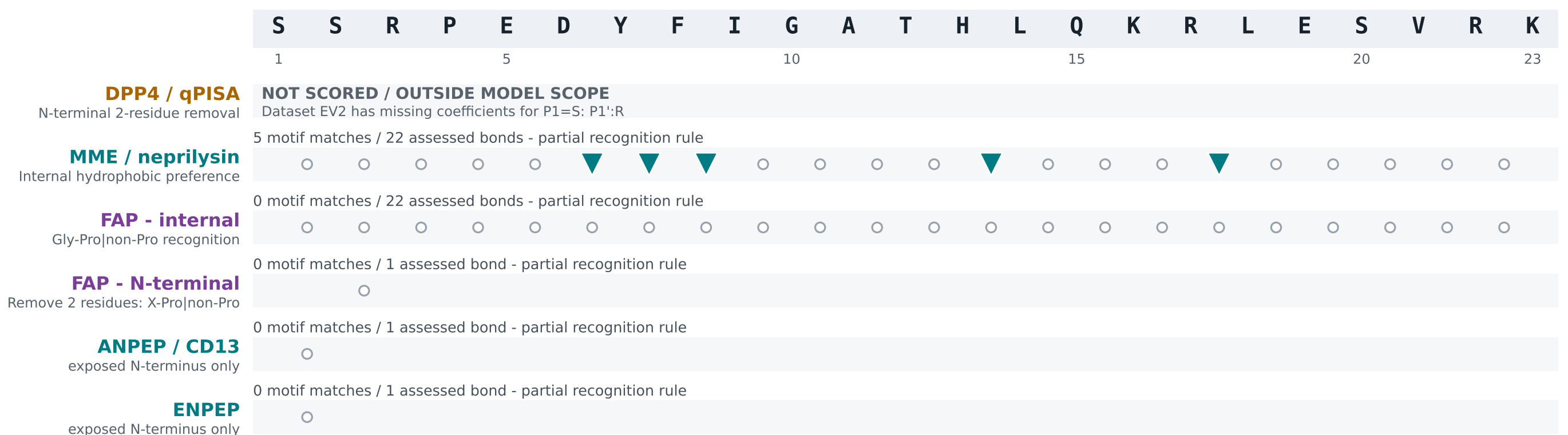


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

DYNC1H1 p.Val314Ile | JLF V3, JLF V2 | 24 aa

disclosed minimal epitope KRFHATISF at 2-10

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

NetChop Cterm (ligand) native 0-1

Pepsickle human (in vivo) native 0-1

NetCleave I native 0-1

NetChop 20S (in vitro) native 0-1

0.5 display

0.5 display

0.5 display

0.5 display

MHC-I: 8 shown / 18 eligible (top-10 cap)

B27:05 1.7%

C01:02 0.69%

C01:02 1.9%

B27:05 1.9%

C07:01 0.0021%

B08:01 1.3%

B27:05 0.92%

A01:01 1.3%

G K R F H A T I S F D T D T G L K Q A L E T K K

DP01:03/04:01 3%

DR03:01 2.2%

DQ05:01/02:01 2.1%

DQ05:01/02:01 1.5%

DR03:01 1.6%

MHC-II: 5 shown / 47 eligible (top-10 cap)

ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure

G K R F H A T I S F D T D T G L K Q A L E T K K

DPP4 / qPISA

N-terminal 2-residue removal

NOT SCORED / OUTSIDE MODEL SCOPE

Dataset EV2 has missing coefficients for P1=K; P1':R

5 motif matches / 23 assessed bonds - partial recognition rule

MME / neprilysin

Internal hydrophobic preference

Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

FAP - internal

Gly-Pro|non-Pro recognition

0 motif matches / 23 assessed bonds - partial recognition rule

FAP - N-terminal

Remove 2 residues: X-Pro|non-Pro

0 motif matches / 1 assessed bond - partial recognition rule

ANPEP / CD13

exposed N-terminus only

0 motif matches / 1 assessed bond - partial recognition rule

ENPEP

exposed N-terminus only

0 motif matches / 1 assessed bond - partial recognition rule

Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

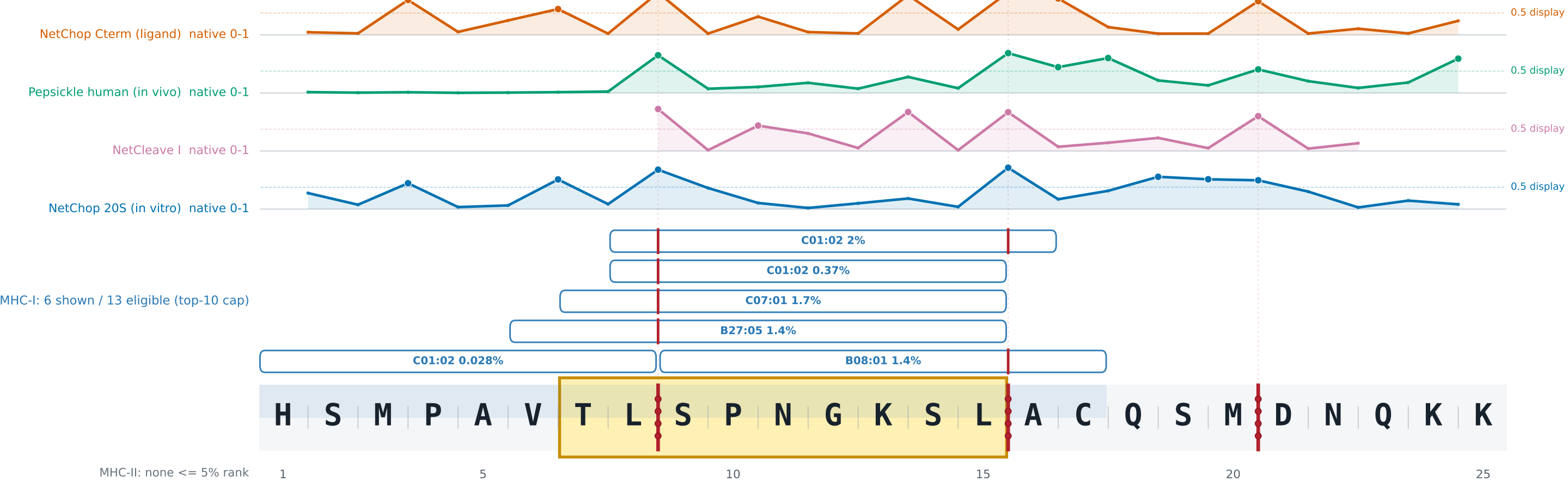
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

CDC40 p.Trp475Ser | JLF V3 | 25 aa

disclosed minimal epitope TLSPNGKSL at 7-15

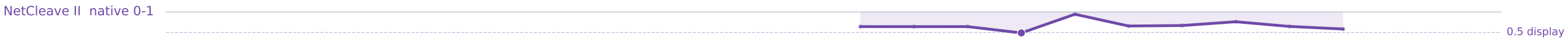
CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

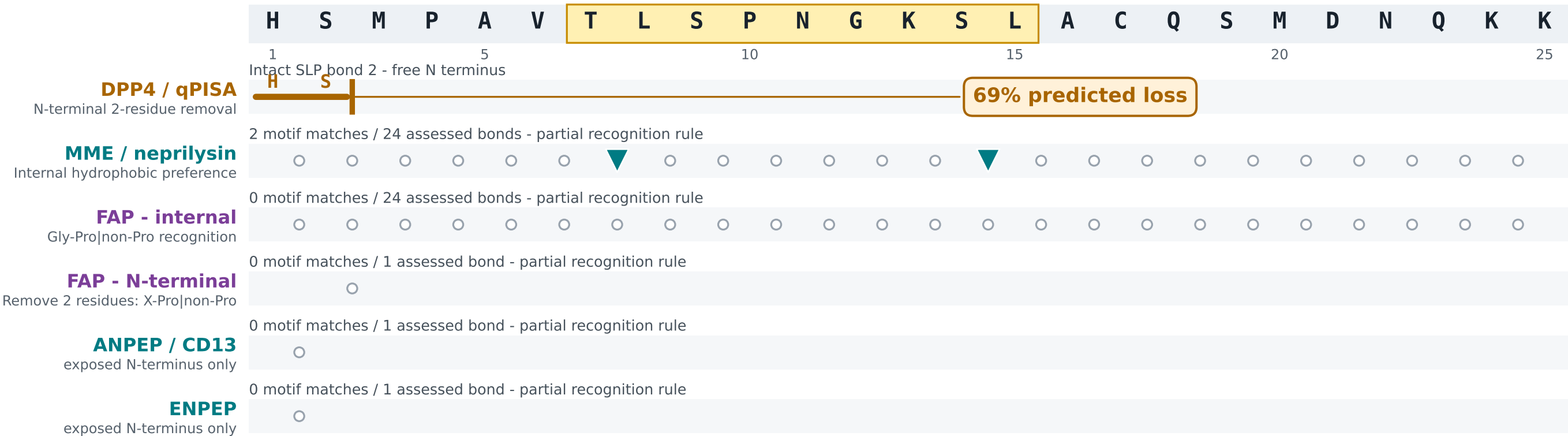


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

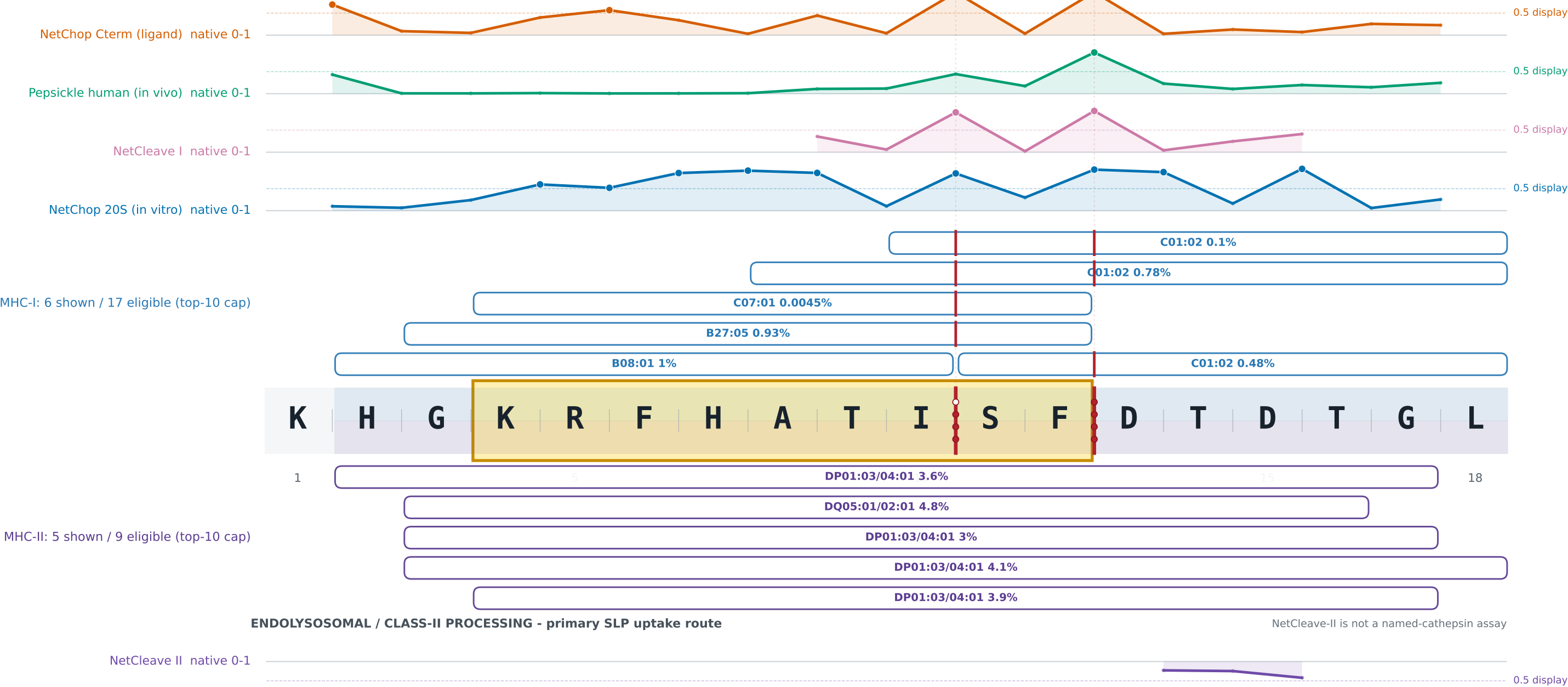
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

DYNC1H1 p.Val314Ile | CeGaT | 18 aa

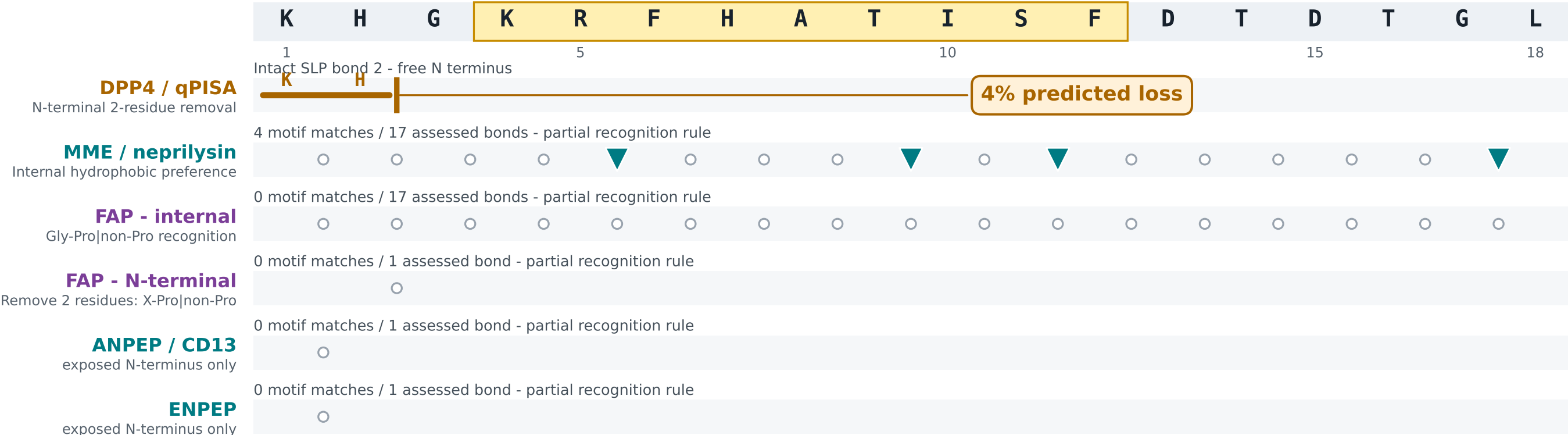
disclosed minimal epitope KRFHATISF at 4-12

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

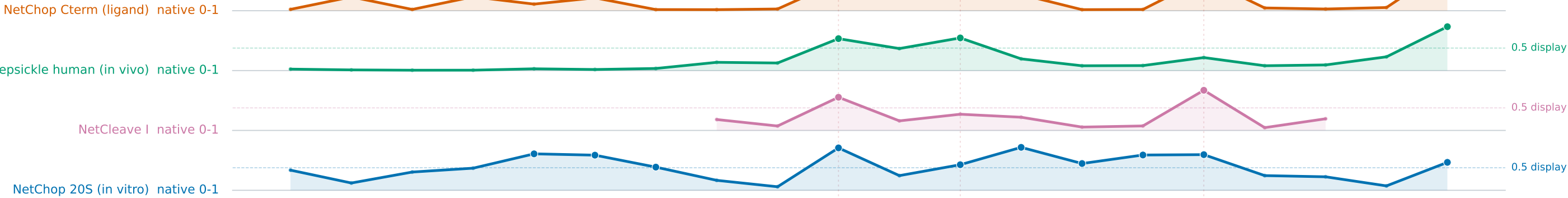
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

SMC5 p.Arg371Lys | JLF V3 | 21 aa

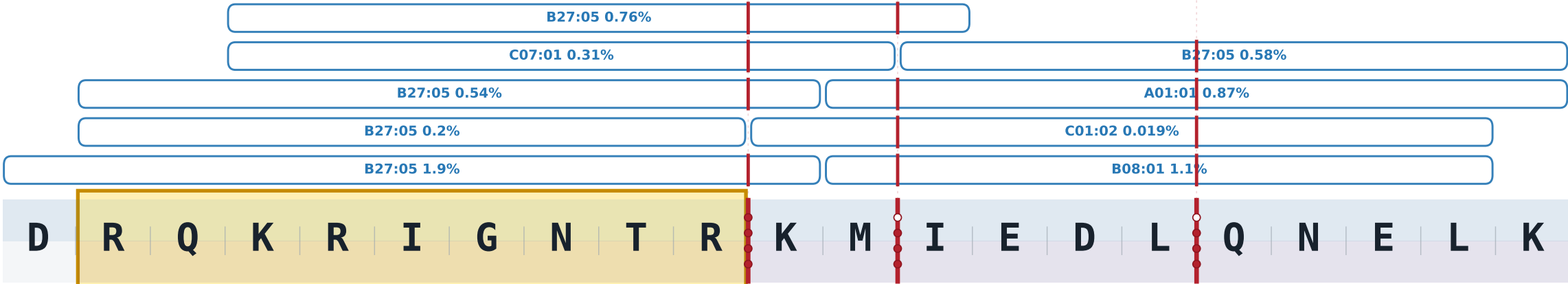
disclosed minimal epitope RQKRIGNTR at 2-10

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

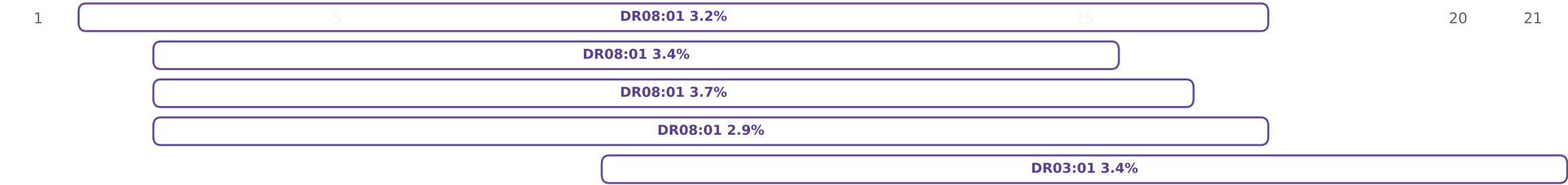
four model-native tracks



MHC-I: 9 shown / 35 eligible (top-10 cap)



MHC-II: 5 shown / 13 eligible (top-10 cap)

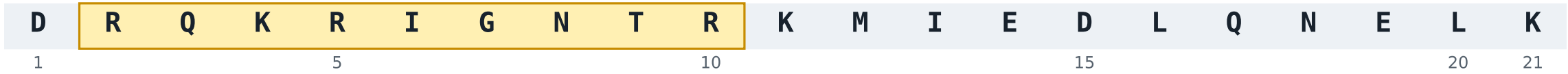


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



DPP4 / qPISA
N-terminal 2-residue removal

NOT SCORED / OUTSIDE MODEL SCOPE
Dataset EV2 has missing coefficients for P1=R: P1':Q

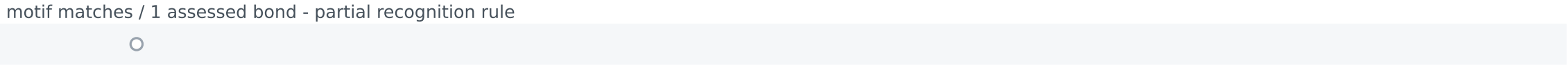
MME / neprilysin
Internal hydrophobic preference



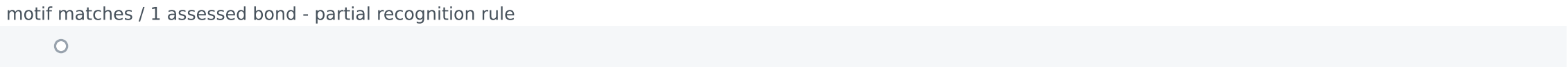
FAP - internal
Gly-Pro|non-Pro recognition



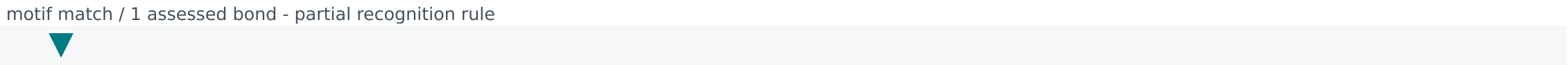
FAP - N-terminal
Remove 2 residues: X-Pro|non-Pro



ANPEP / CD13
exposed N-terminus only



ENPEP
exposed N-terminus only



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

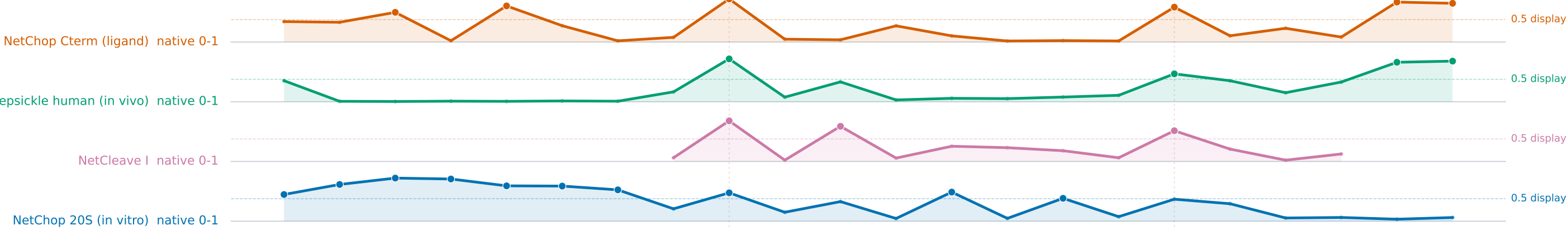
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

CUL9 p.Asp403Tyr | JLF V2 | 23 aa

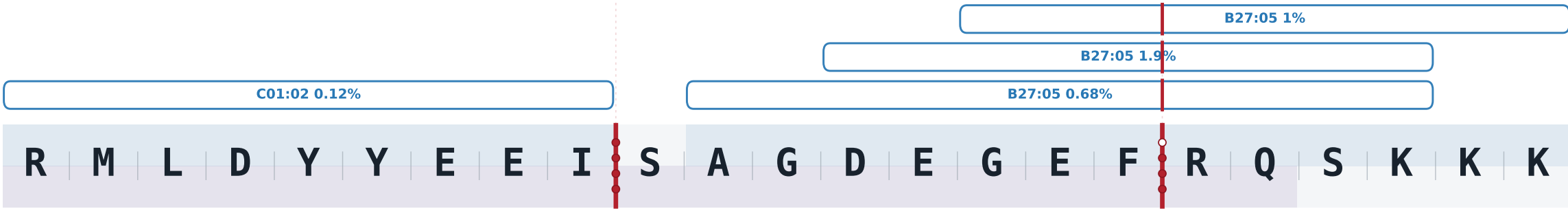
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

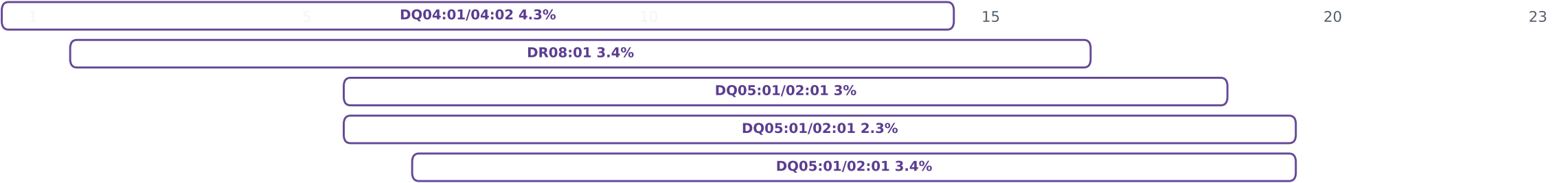
four model-native tracks



MHC-I: 4 shown / 7 eligible (top-10 cap)



R M L D Y Y E E I S A G D E G E F R Q S K K K



MHC-II: 5 shown / 13 eligible (top-10 cap)

ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure

R M L D Y Y E E I S A G D E G E F R Q S K K K

1 5 10 15 20 23

DPP4 / qPISA

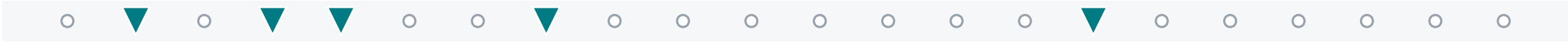
N-terminal 2-residue removal

NOT SCORED / OUTSIDE MODEL SCOPE
Dataset EV2 has missing coefficients for P1=M: P2:R

MME / neprilysin

Internal hydrophobic preference

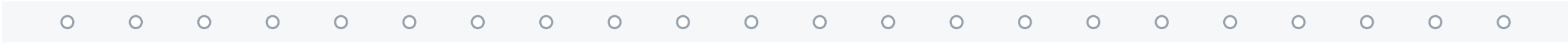
5 motif matches / 22 assessed bonds - partial recognition rule



FAP - internal

Gly-Pro|non-Pro recognition

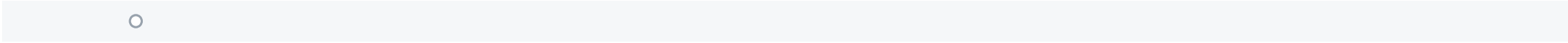
0 motif matches / 22 assessed bonds - partial recognition rule



FAP - N-terminal

Remove 2 residues: X-Pro|non-Pro

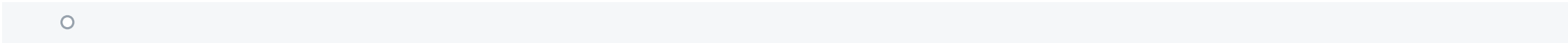
0 motif matches / 1 assessed bond - partial recognition rule



ANPEP / CD13

exposed N-terminus only

0 motif matches / 1 assessed bond - partial recognition rule



ENPEP

exposed N-terminus only

0 motif matches / 1 assessed bond - partial recognition rule



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

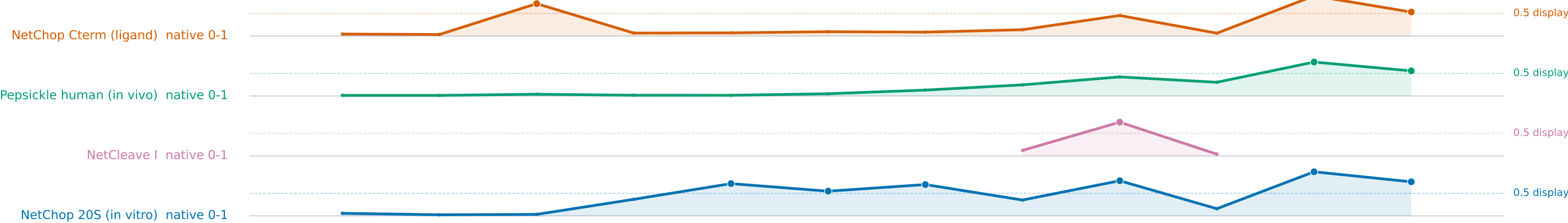
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

NME1 p.Gly24Arg | CeGaT | 13 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



MHC-I: 1 shown / 1 eligible (top-10 cap)

C01:02 1.1%



MHC-II: none <= 5% rank

1

5

10

13

ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure

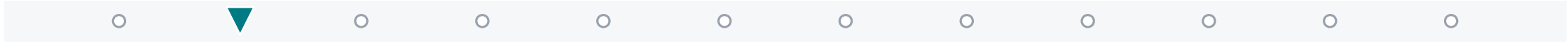


DPP4 / qPISA

N-terminal 2-residue removal

NOT SCORED / OUTSIDE MODEL SCOPE
Dataset EV2 has missing coefficients for P1=P: P2:P

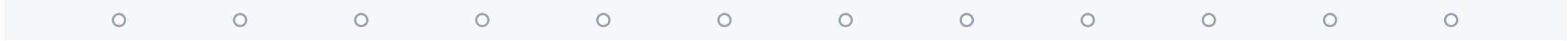
1 motif match / 12 assessed bonds - partial recognition rule



MME / neprilysin

Internal hydrophobic preference

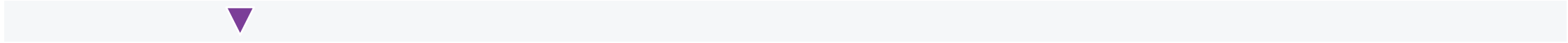
0 motif matches / 12 assessed bonds - partial recognition rule



FAP - internal

Gly-Pro|non-Pro recognition

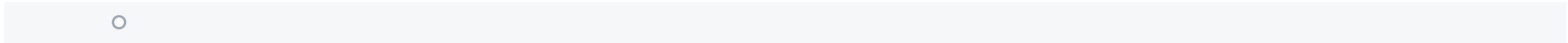
1 motif match / 1 assessed bond - partial recognition rule



FAP - N-terminal

Remove 2 residues: X-Pro|non-Pro

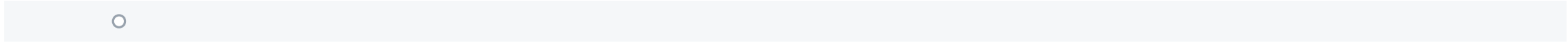
0 motif matches / 1 assessed bond - partial recognition rule



ANPEP / CD13

exposed N-terminus only

0 motif matches / 1 assessed bond - partial recognition rule



ENPEP

exposed N-terminus only

Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

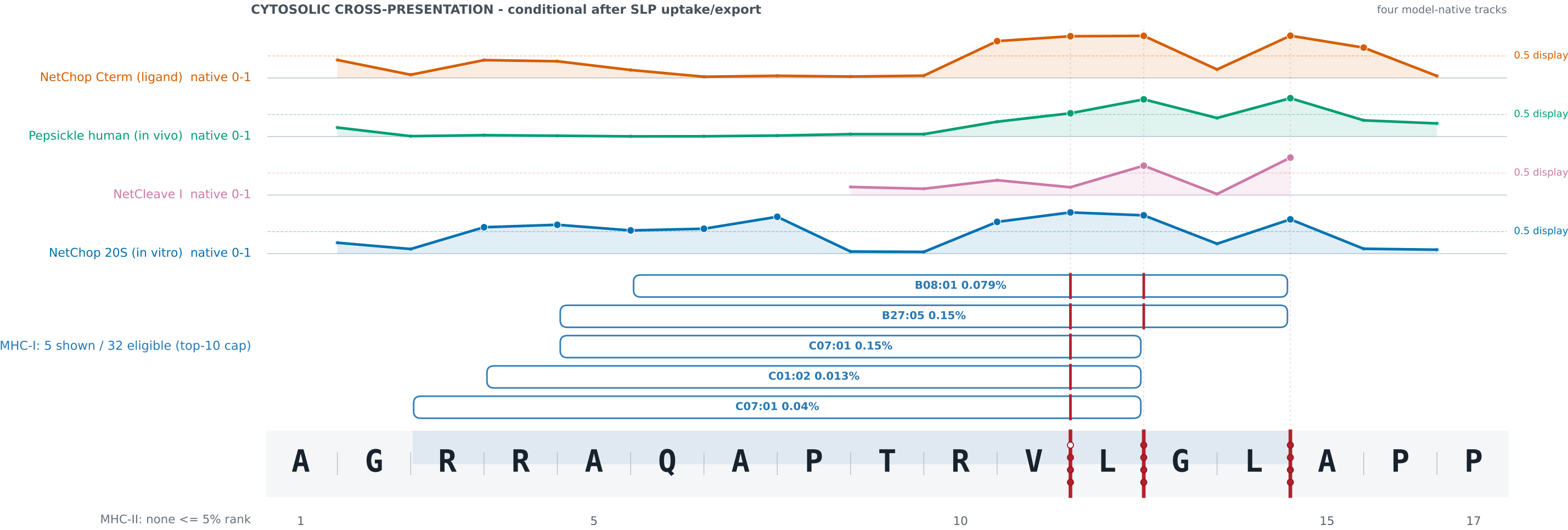
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

MYO15B fs | JLF V2, JLF V1 | 17 aa

minimal epitope/window not disclosed for this SLP

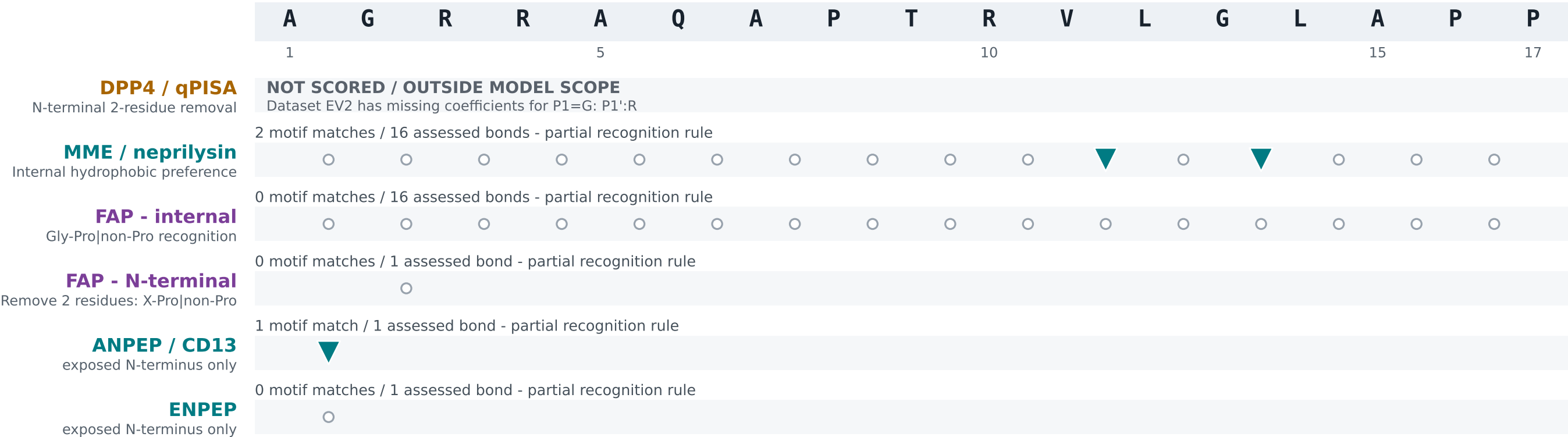


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

VPS72 p.Lys115Arg | JLF V3 | 27 aa

disclosed minimal epitope ERALLPLEL at 9-17

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

NetChop Cterm (ligand) native 0-1

Pepsickle human (in vivo) native 0-1

NetCleave I native 0-1

NetChop 20S (in vitro) native 0-1

0.5 display

0.5 display

0.5 display

0.5 display

MHC-I: 5 shown / 12 eligible (top-10 cap)

MHC-II: 5 shown / 10 eligible (top-10 cap)

G | S | S | Q | K | A | R | E | E | R | A | L | L | P | L | E | L | Q | D | D | G | S | D | S | R | K | S

1

5

10

15

25

27

ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure

G | S | S | Q | K | A | R | E | E | R | A | L | L | P | L | E | L | Q | D | D | G | S | D | S | R | K | S

1

5

10

15

20

25

27

DPP4 / qPISA

N-terminal 2-residue removal

21% predicted loss

MME / neprilysin

Internal hydrophobic preference

FAP - internal

Gly-Pro|non-Pro recognition

FAP - N-terminal

Remove 2 residues: X-Pro|non-Pro

ANPEP / CD13

exposed N-terminus only

ENPEP

exposed N-terminus only

Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

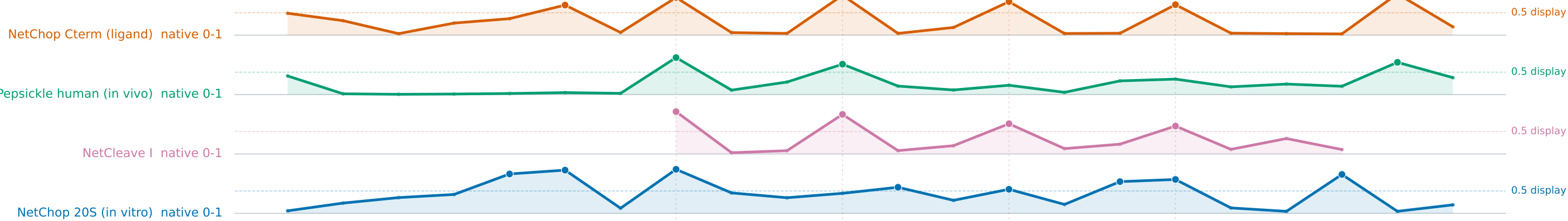
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

EXOC4 p.Ser34Ile | JLF V3, JLF V2 | 23 aa

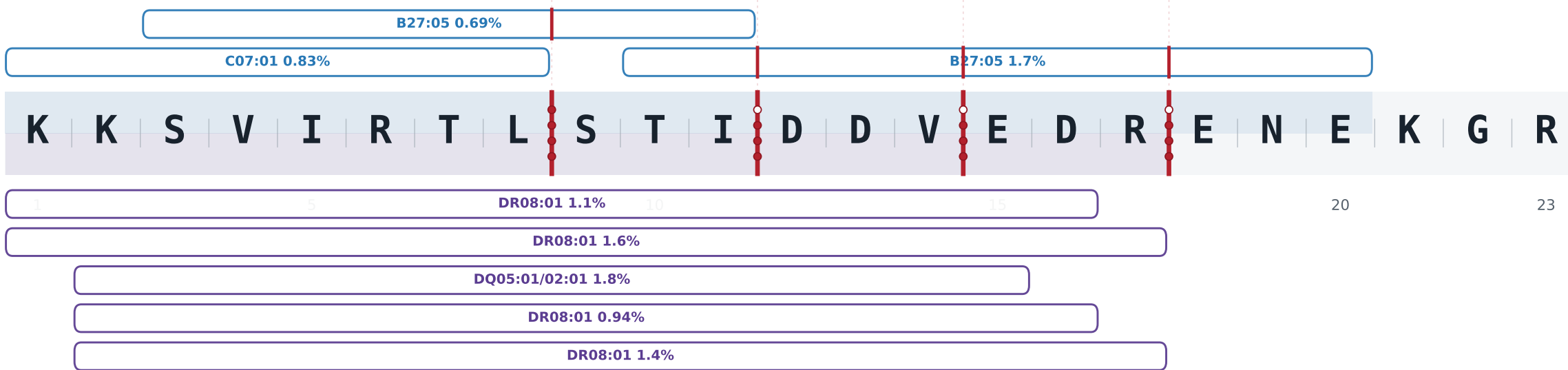
minimal epitope/window not disclosed for this SLP



four model-native tracks



MHC-I: 3 shown / 4 eligible (top-10 cap)



MHC-II: 5 shown / 38 eligible (top-10 cap)

ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



DPP4 / qPISA

N-terminal 2-residue removal

NOT SCORED / OUTSIDE MODEL SCOPE

Dataset EV2 has missing coefficients for $P1=K$: $P1':S$

3 motif matches / 22 assessed bonds - partial recognition rule



0 motif matches / 22 assessed bonds - partial recognition rule



0 motif matches / 1 assessed bond - partial recognition rule

0 motif matches / 1 assessed bond - partial recognition rule

0 motif matches / 1 assessed bond - partial recognition rule

Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

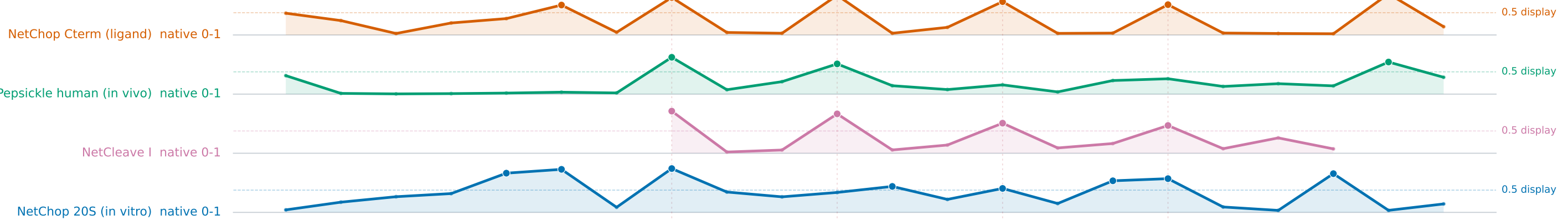
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

EPG5 p.Ala869Thr | CeGaT | 23 aa

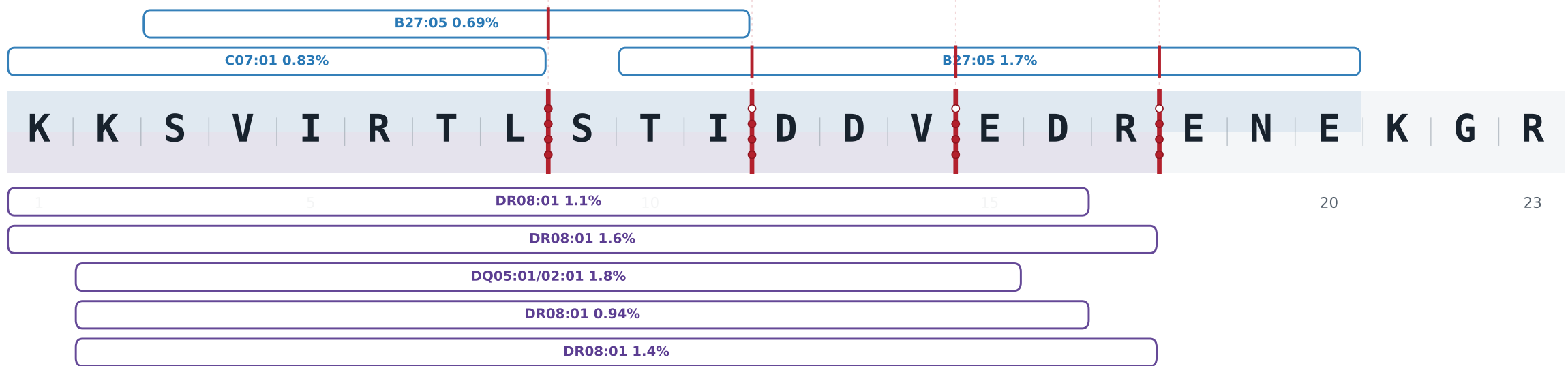
minimal epitope/window not disclosed for this SLP



four model-native tracks



MHC-I: 3 shown / 4 eligible (top-10 cap)



MHC-II: 5 shown / 38 eligible (top-10 cap)

ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



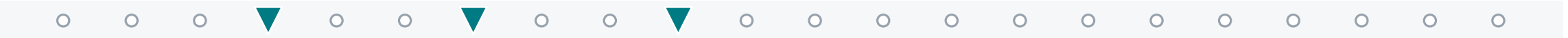
DPP4 / qPISA

N-terminal 2-residue removal

NOT SCORED / OUTSIDE MODEL SCOPE

Dataset EV2 has missing coefficients for P1=K: P1':S

3 motif matches / 22 assessed bonds - partial recognition rule



0 motif matches / 22 assessed bonds - partial recognition rule



0 motif matches / 1 assessed bond - partial recognition rule

0 motif matches / 1 assessed bond - partial recognition rule

0 motif matches / 1 assessed bond - partial recognition rule

Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

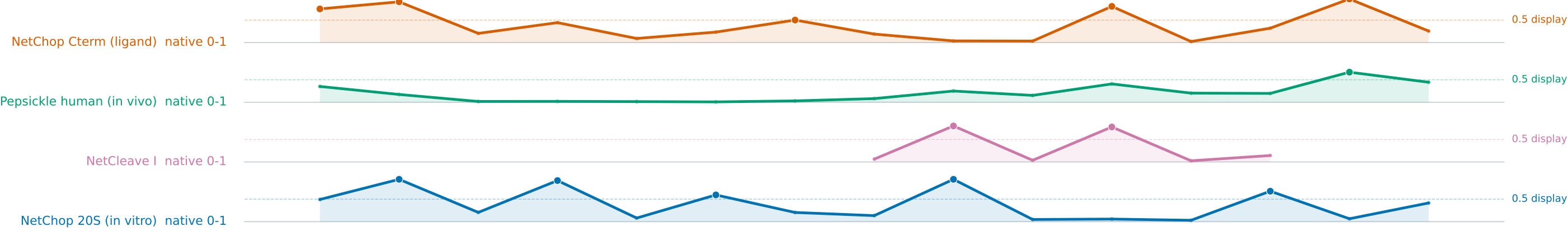
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

TECPR1 p.Thr259fs | CeGaT | 16 aa

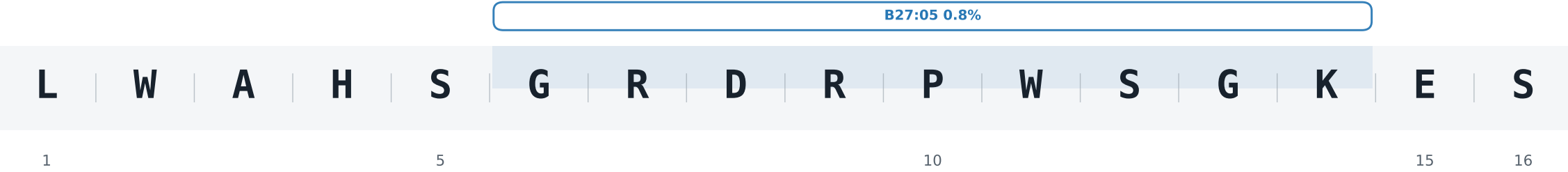
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



MHC-II: none <= 5% rank

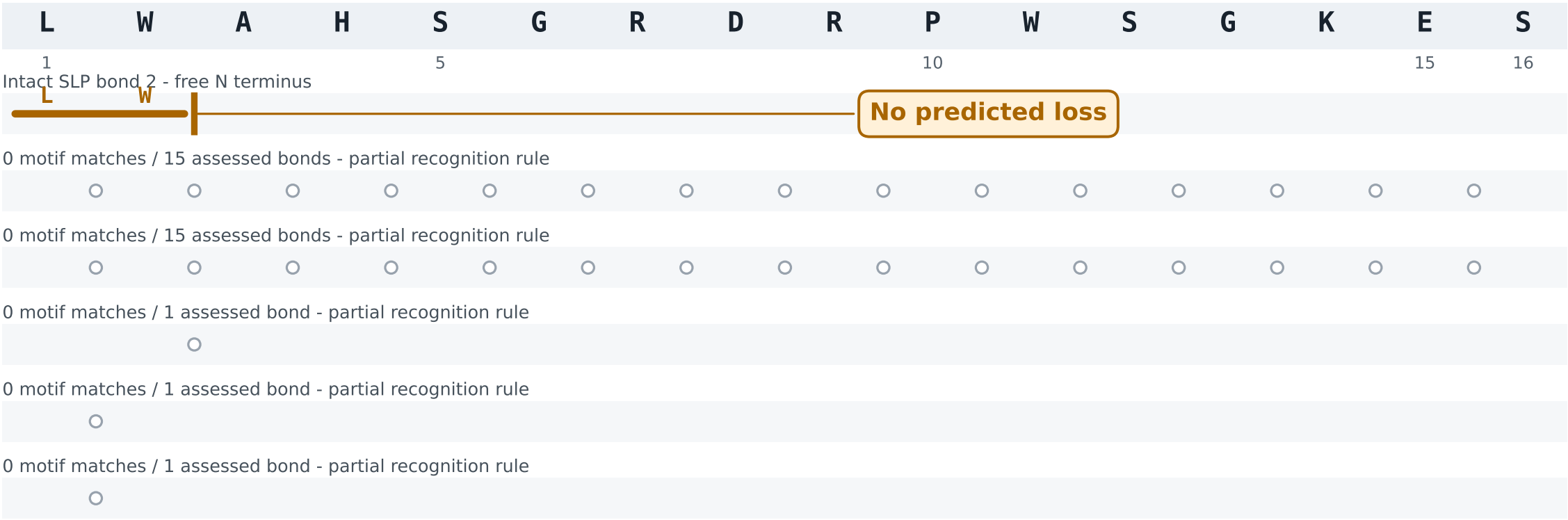


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

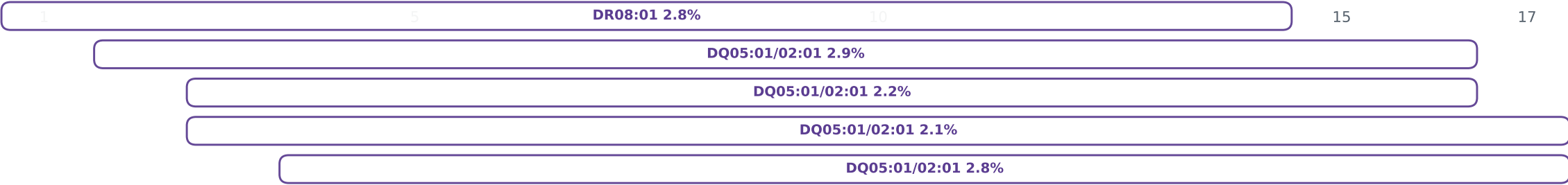
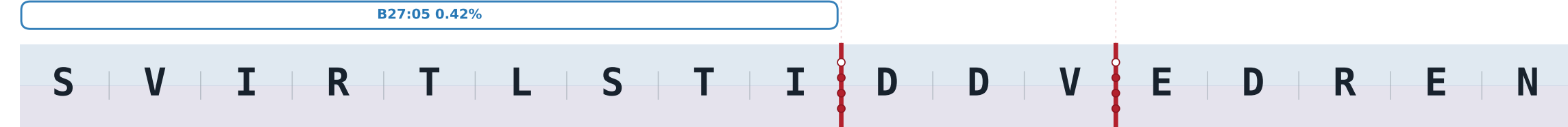
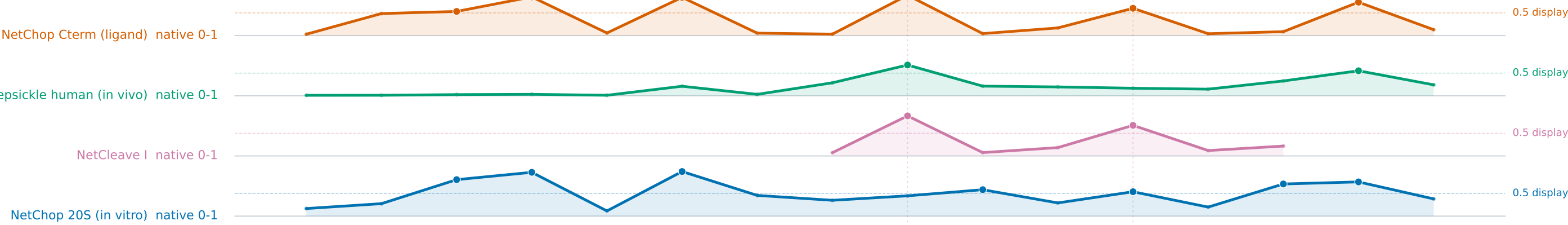
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

EXOC4 p.Ser34Ile | CeGaT | 17 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

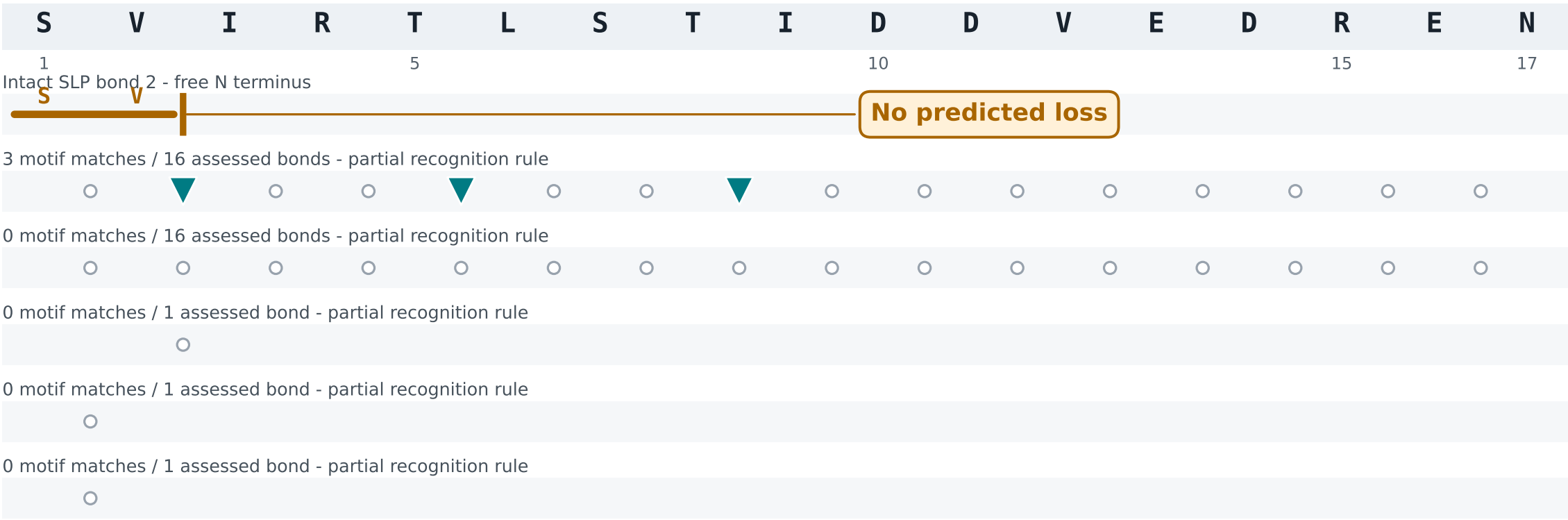


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

SLC25A12 p.Ile347Thr | JLF V2, JLF V1 | 26 aa

disclosed minimal epitope ATAVYPTDL at 9-17

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

NetChop Cterm (ligand) native 0-1

Pepsickle human (in vivo) native 0-1

NetCleave I native 0-1

NetChop 20S (in vitro) native 0-1

0.5 display

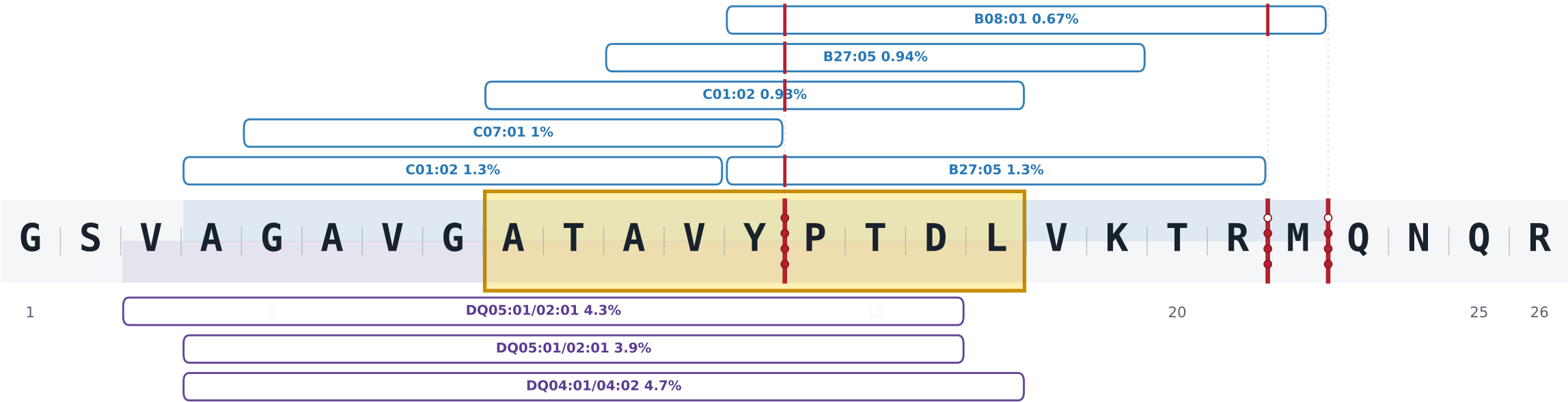
0.5 display

0.5 display

0.5 display

MHC-I: 6 shown / 18 eligible (top-10 cap)

MHC-II: 3 shown / 6 eligible (top-10 cap)



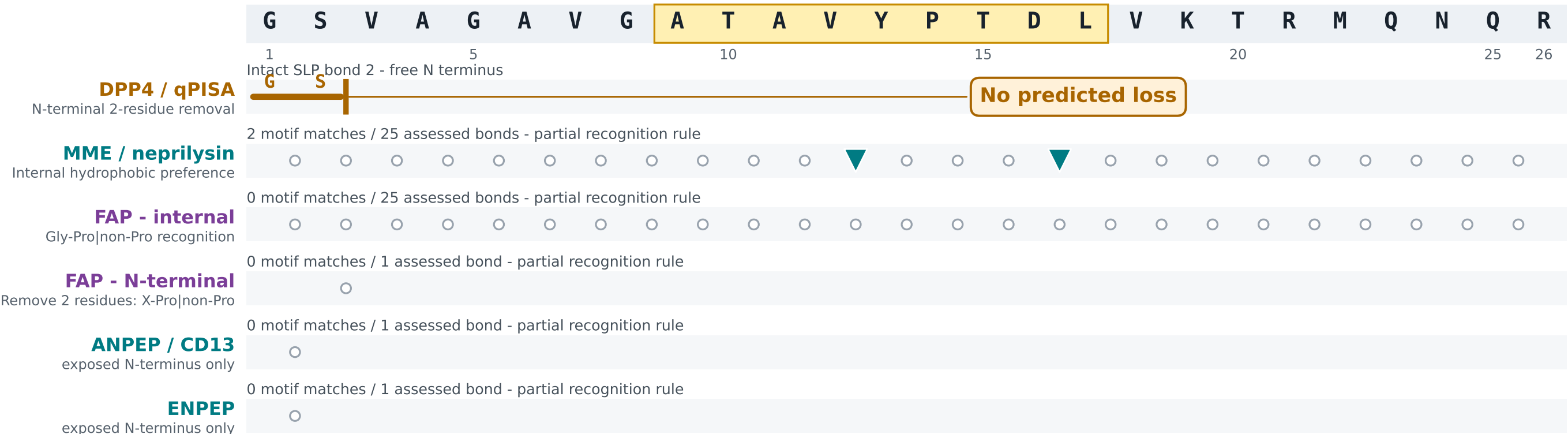
ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

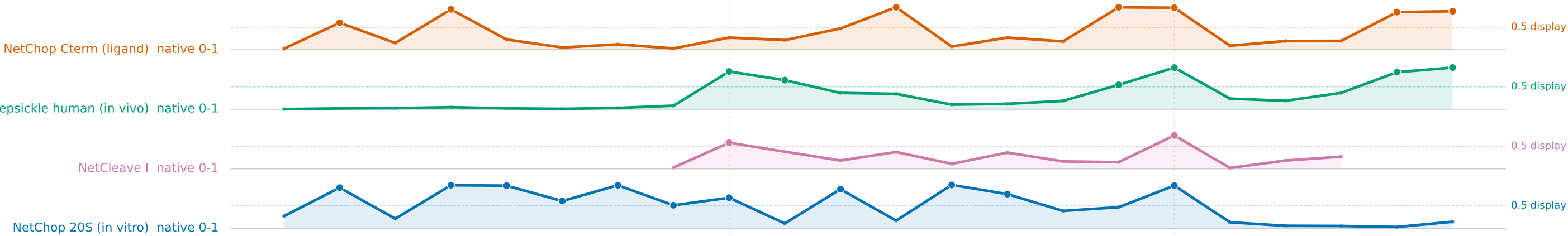
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

PTH1R p.Ile237Phe | JLF V3 | 23 aa

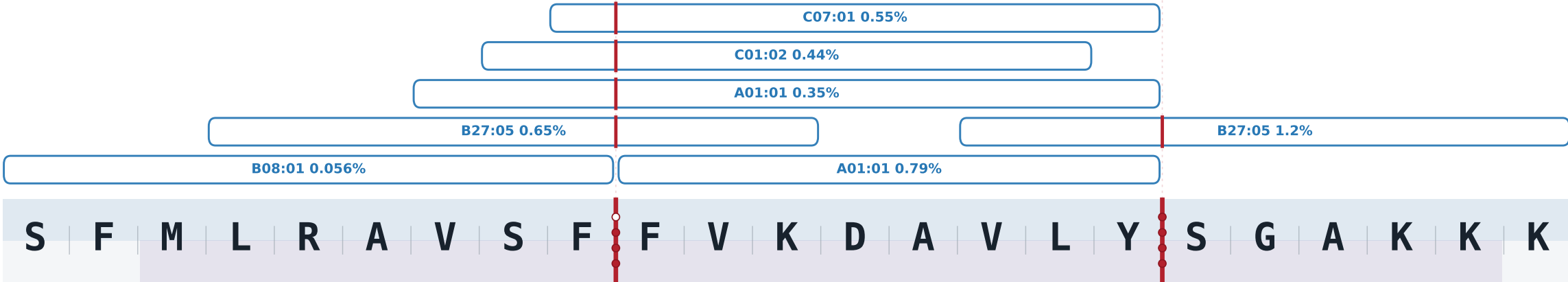
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

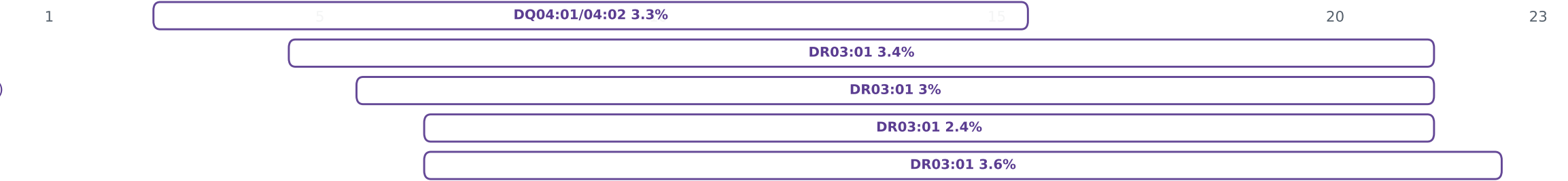
four model-native tracks



MHC-I: 7 shown / 23 eligible (top-10 cap)



MHC-II: 5 shown / 15 eligible (top-10 cap)

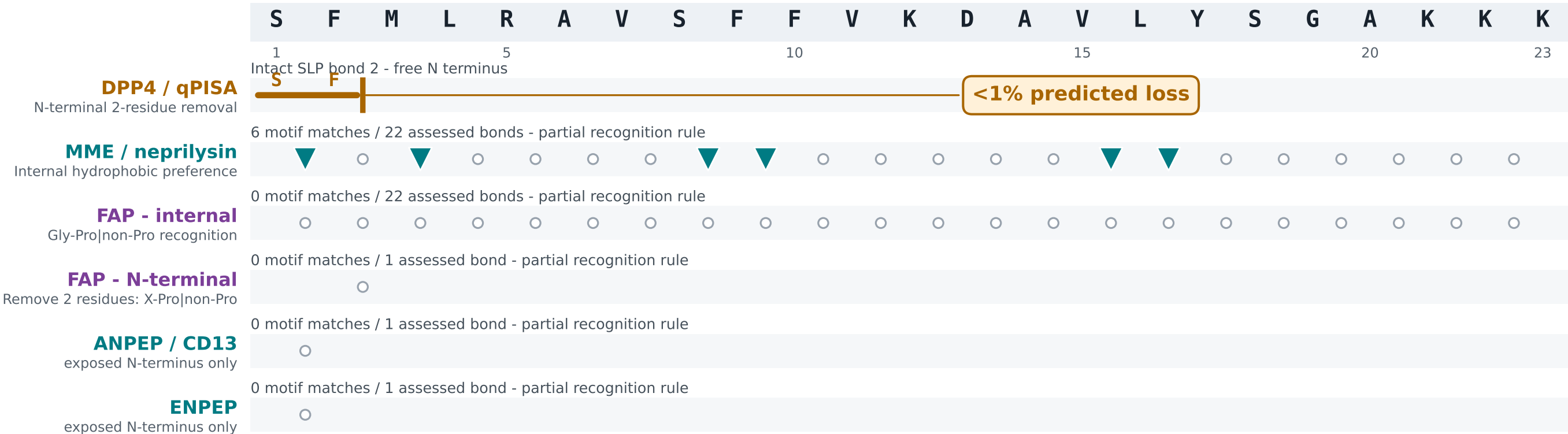


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

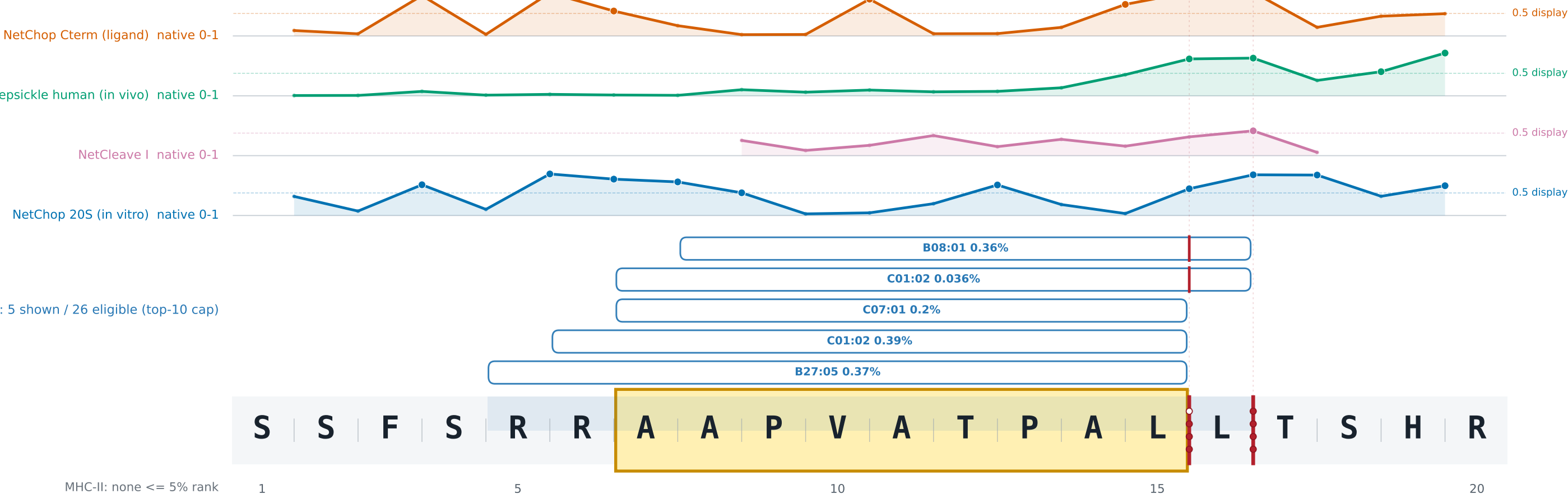
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

PIP5K1A p.Gly474fs | JLF V3 | 20 aa

disclosed minimal epitope AAPVATPAL at 7-15

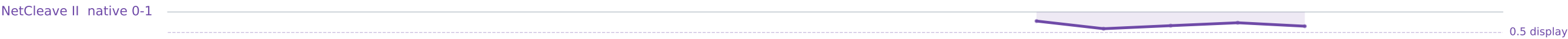
CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

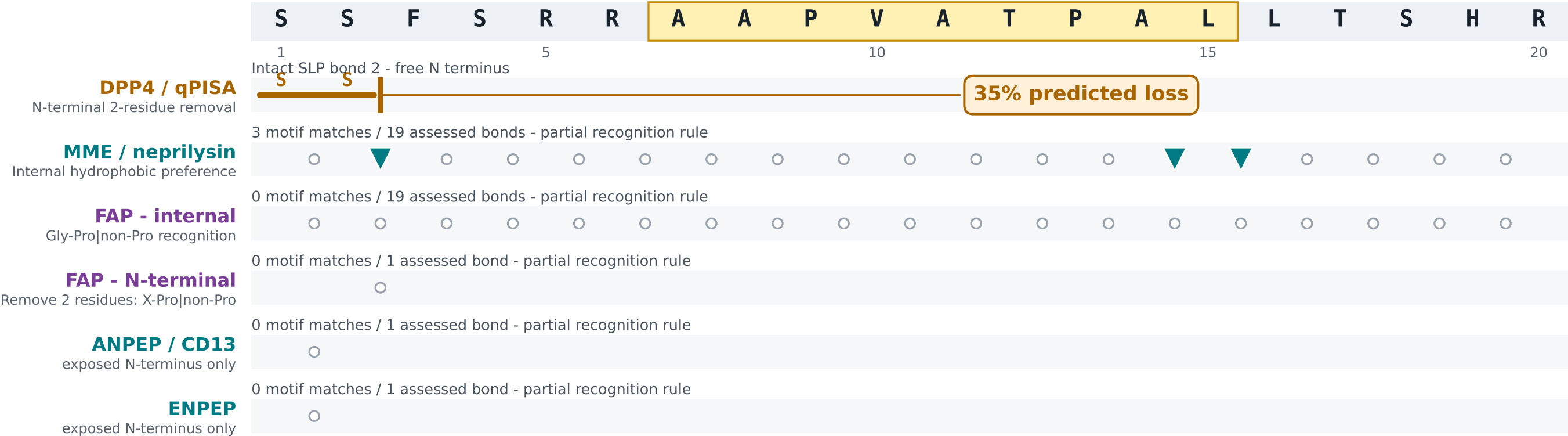


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

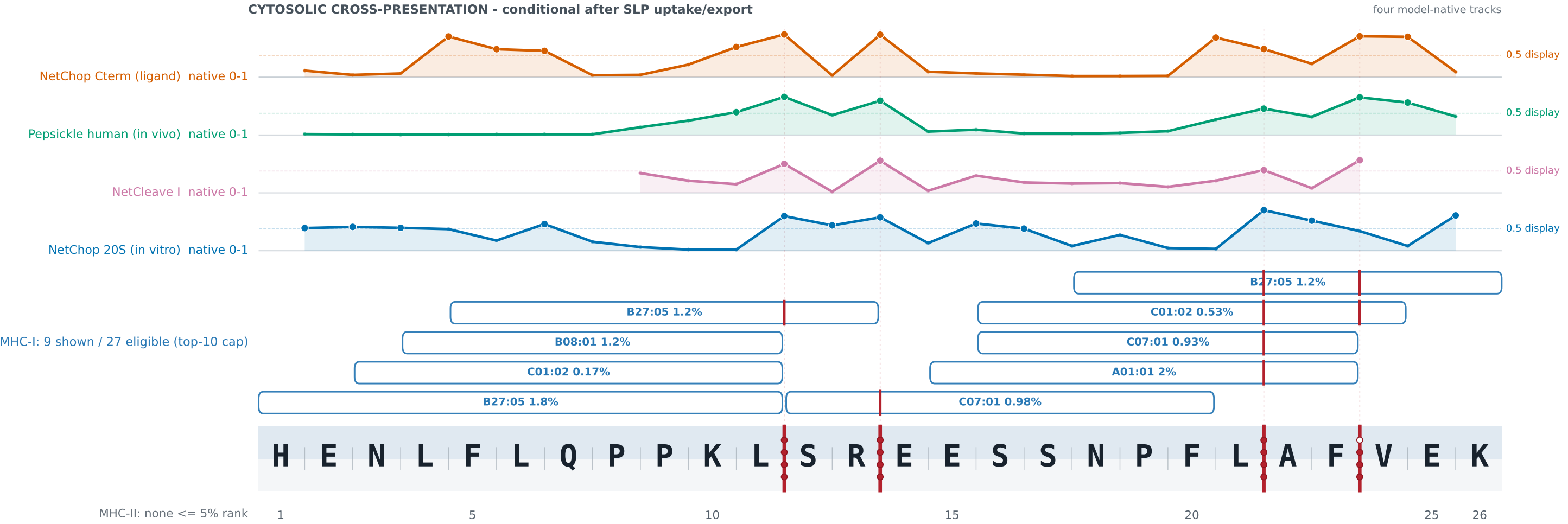
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

KDM3B p.Pro633Ser | JLF V2, JLF V1 | 26 aa

minimal epitope/window not disclosed for this SLP

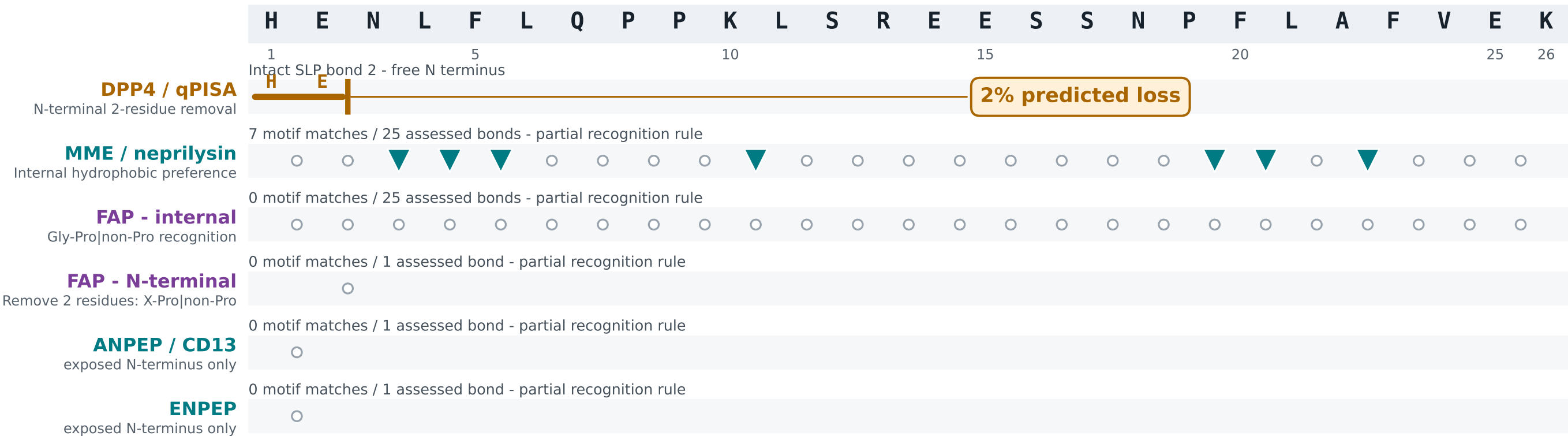


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

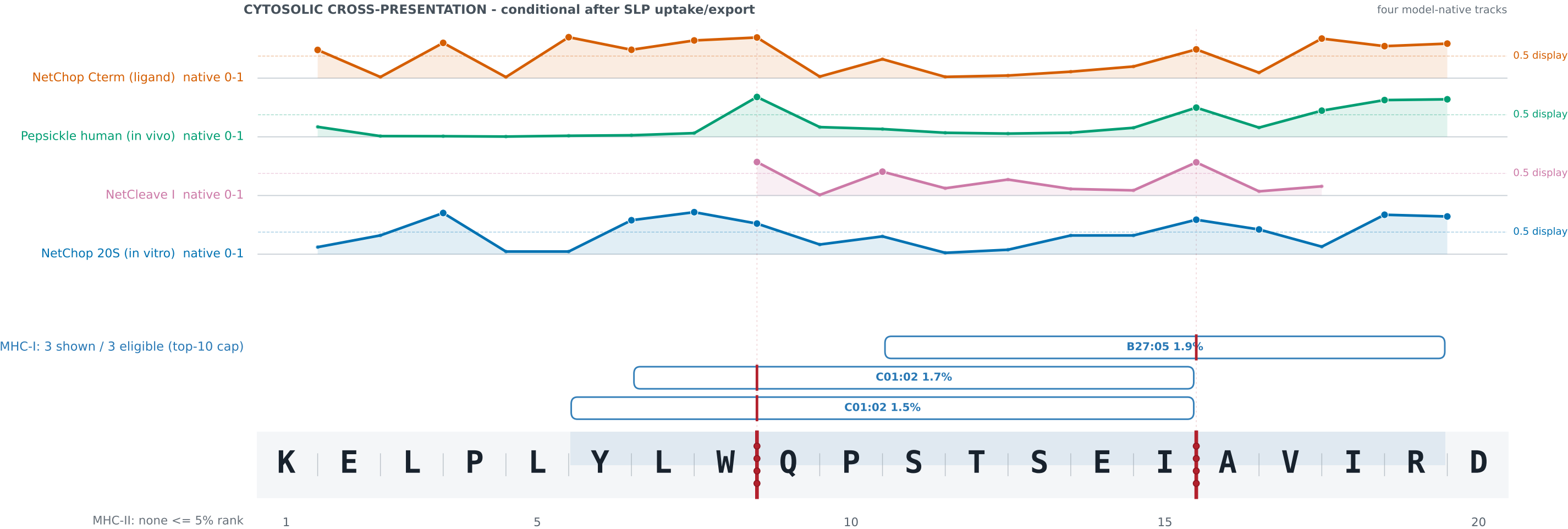
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

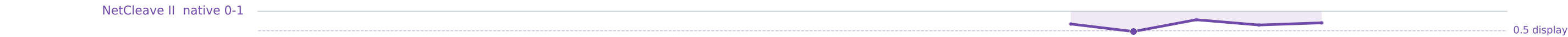
EPG5 p.Ala869Thr | JLF V1 | 20 aa

minimal epitope/window not disclosed for this SLP

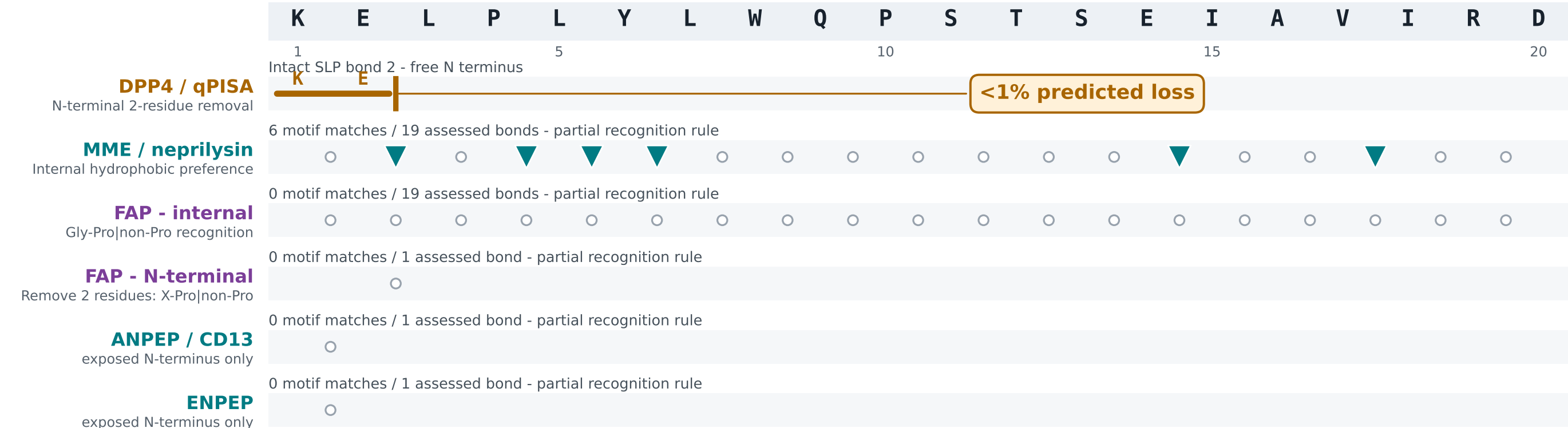


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

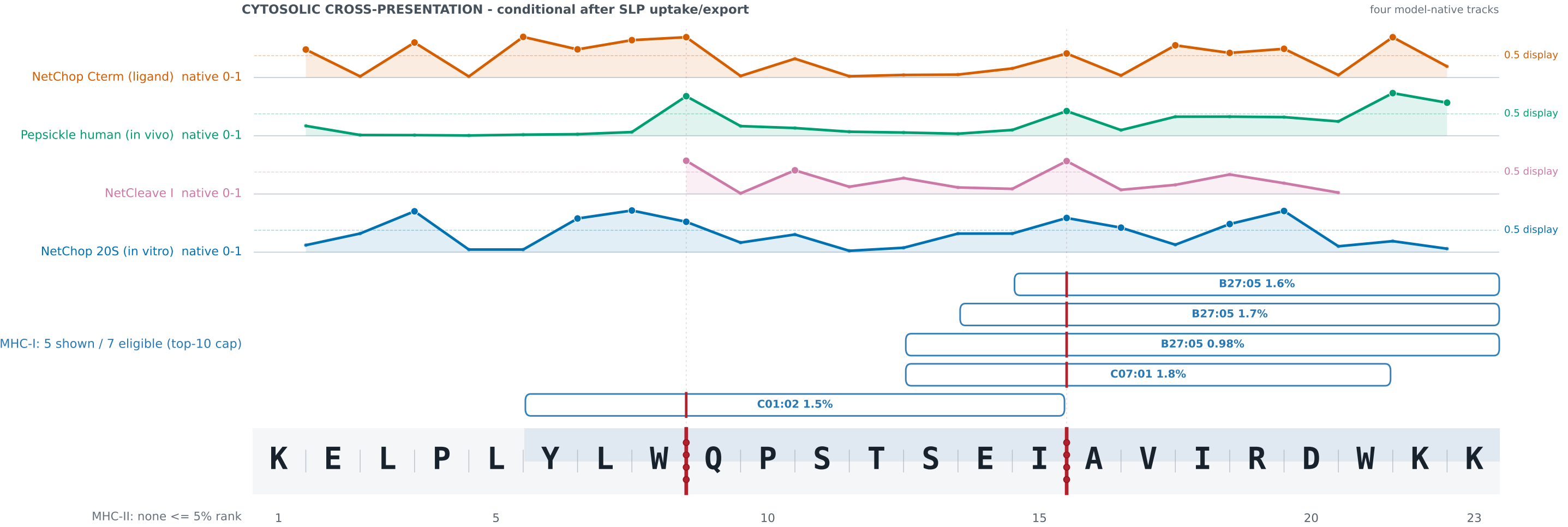
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

EPG5 p.Ala869Thr | JLF V2 | 23 aa

minimal epitope/window not disclosed for this SLP

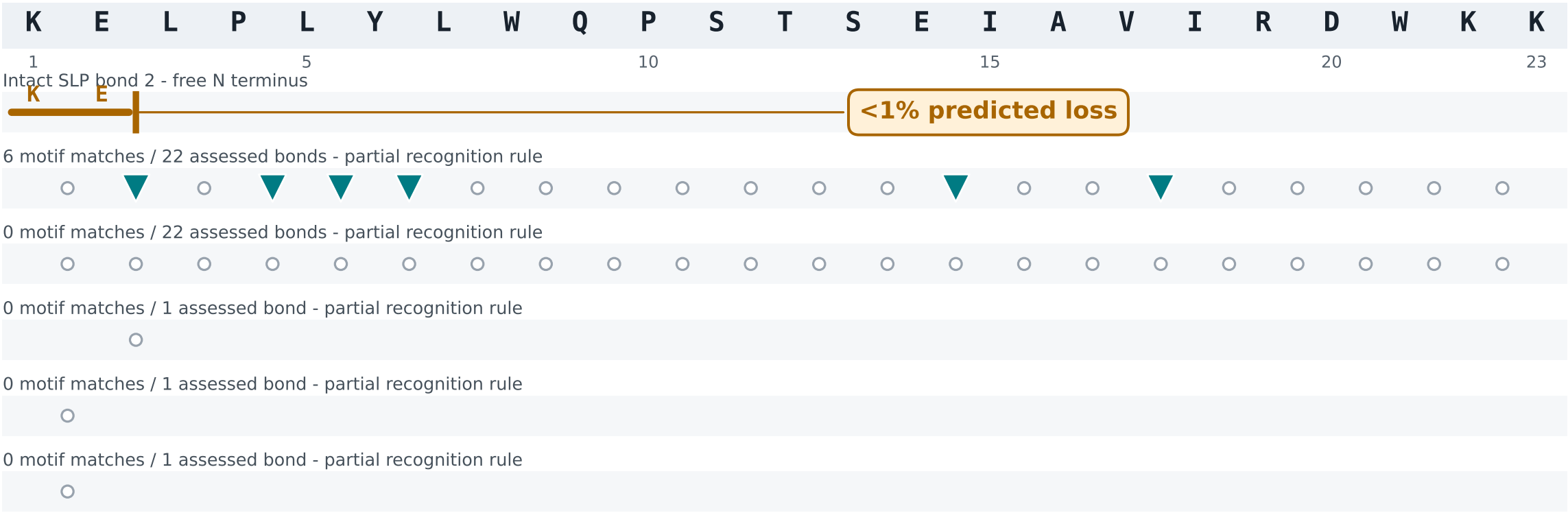


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

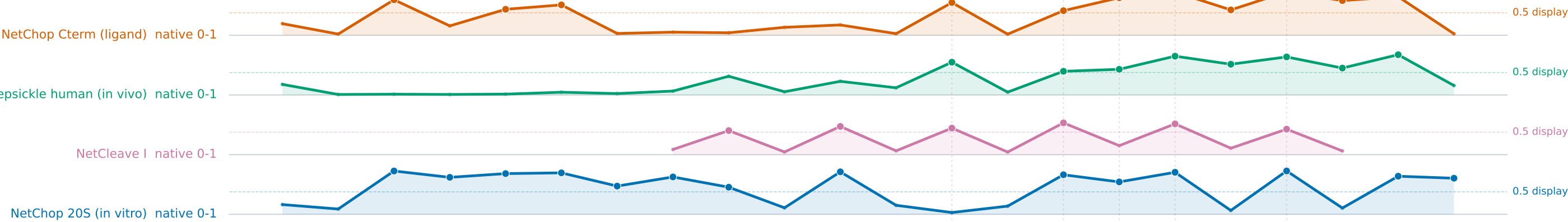
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

MAP2 p.Leu867fs | JLF V3 | 23 aa

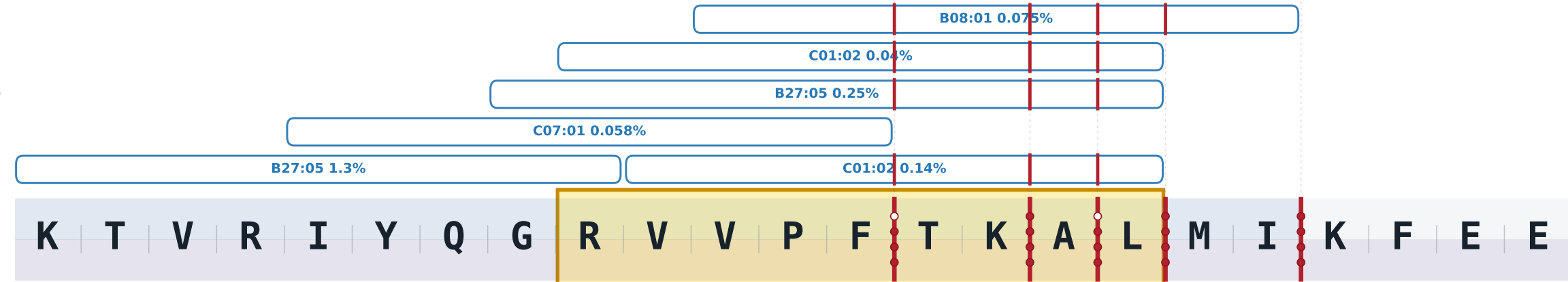
disclosed minimal epitope RVVPFTKAL at 9-17

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



MHC-I: 6 shown / 35 eligible (top-10 cap)

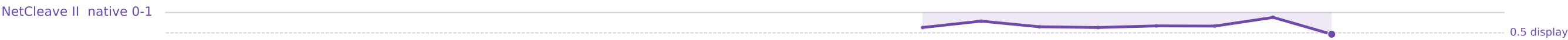


MHC-II: 5 shown / 11 eligible (top-10 cap)

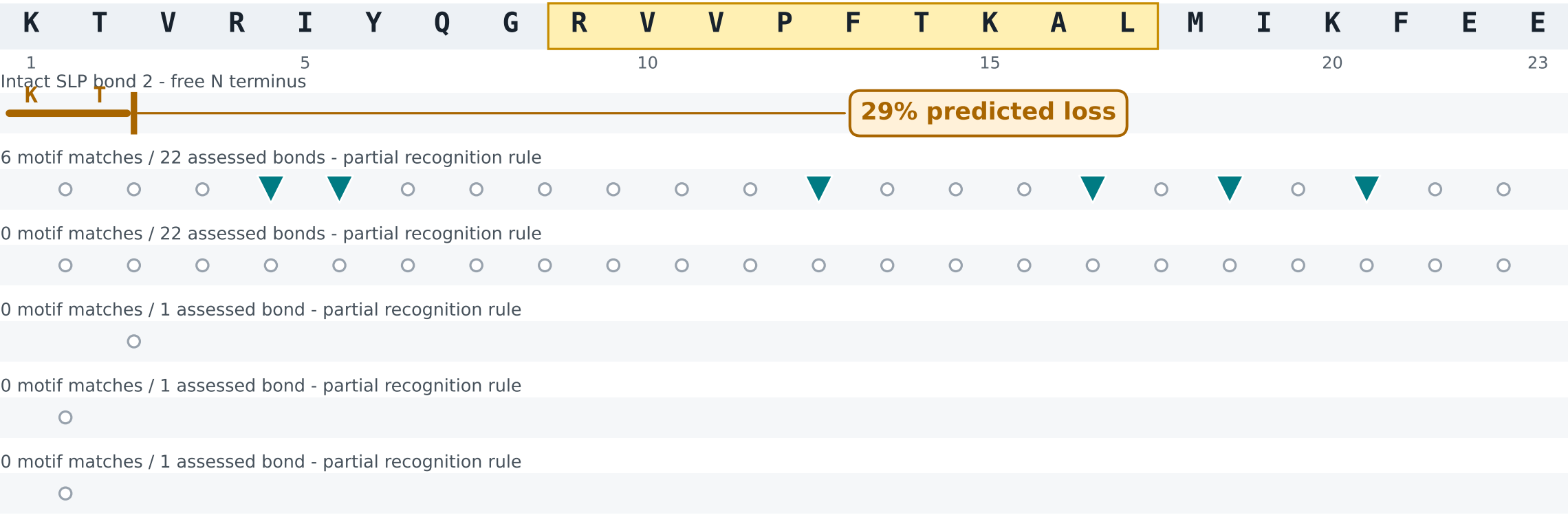


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

MAP2 p.Leu867fs | JLF V2, JLF V1 | 24 aa

disclosed minimal epitope RVVPFTKAL at 9-17

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

NetChop Cterm (ligand) native 0-1

Pepsickle human (in vivo) native 0-1

NetCleave I native 0-1

NetChop 20S (in vitro) native 0-1

0.5 display

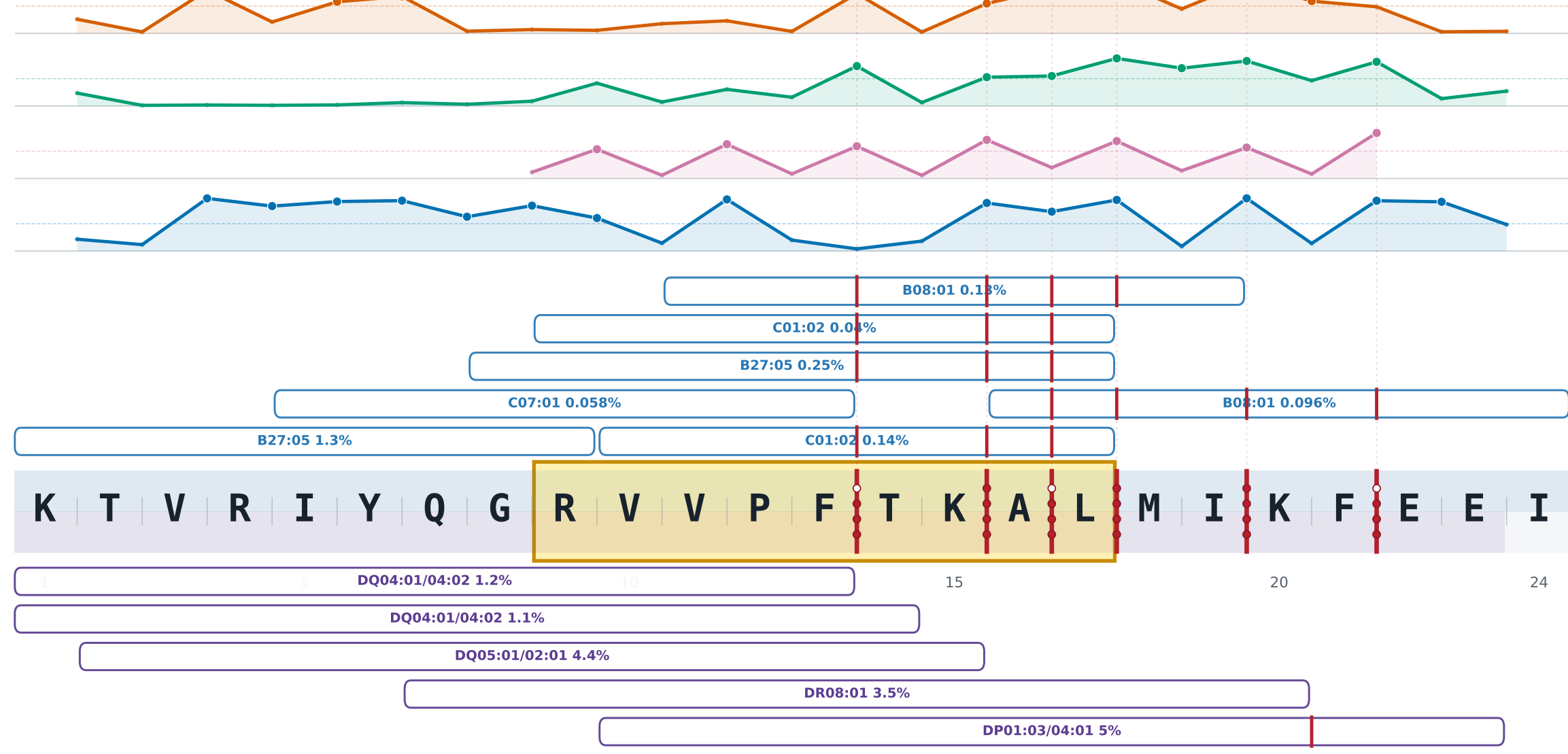
0.5 display

0.5 display

0.5 display

MHC-I: 7 shown / 39 eligible (top-10 cap)

MHC-II: 5 shown / 11 eligible (top-10 cap)



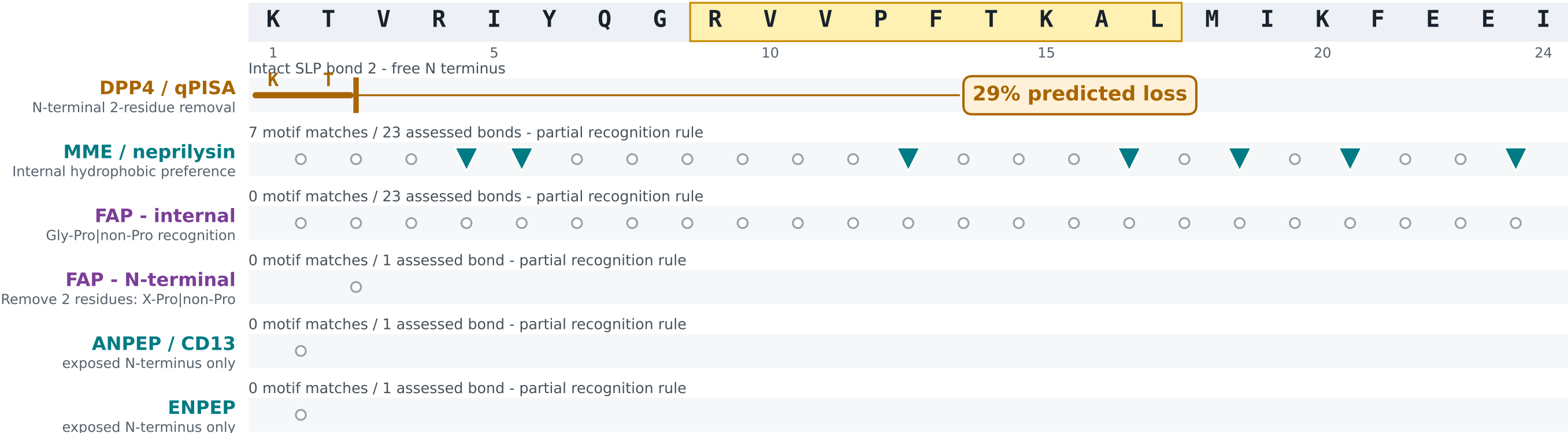
ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

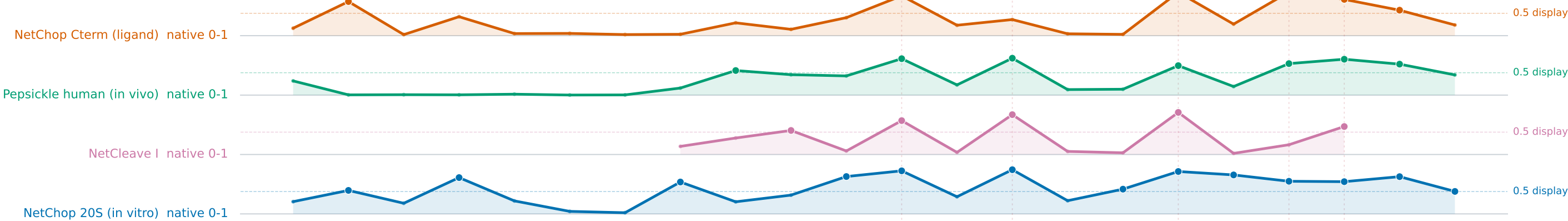
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

BMP1 p.Ser366Thr | JLF V3 | 23 aa

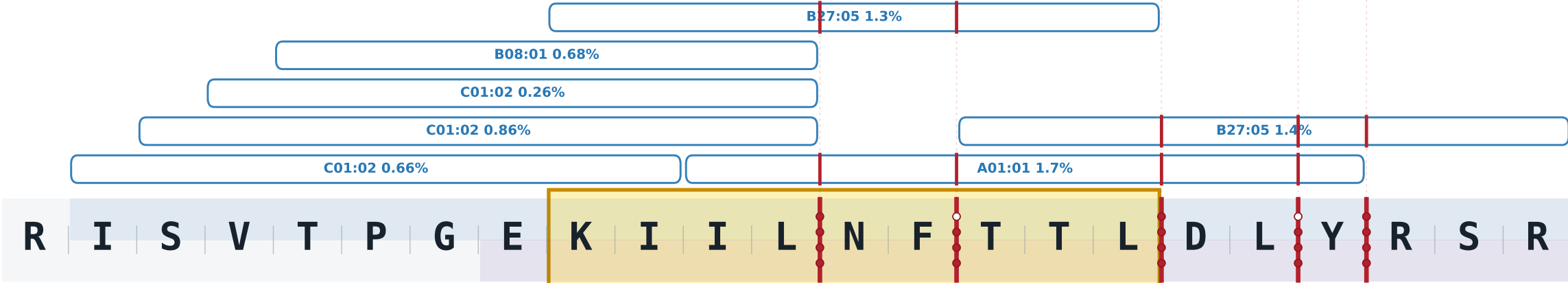
disclosed minimal epitope KIILNFTTL at 9-17

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

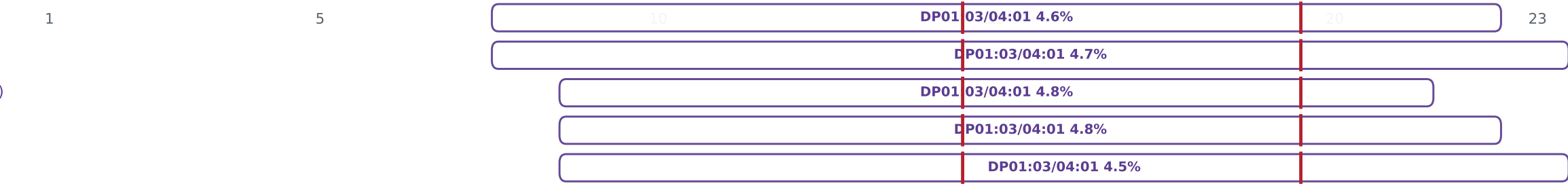
four model-native tracks



MHC-I: 7 shown / 16 eligible (top-10 cap)



MHC-II: 5 shown / 6 eligible (top-10 cap)

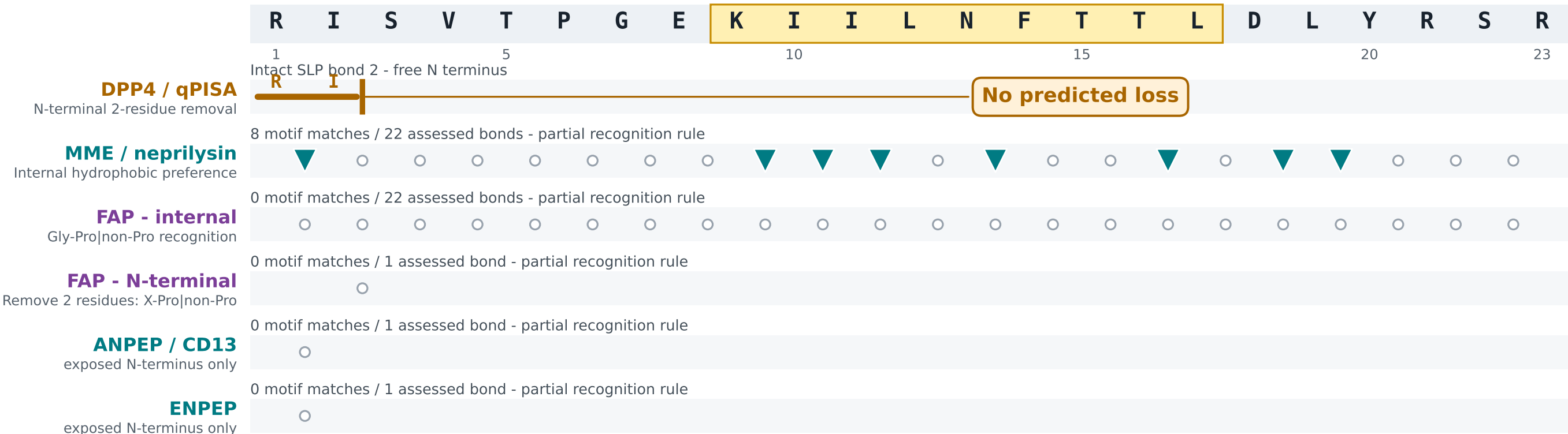


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

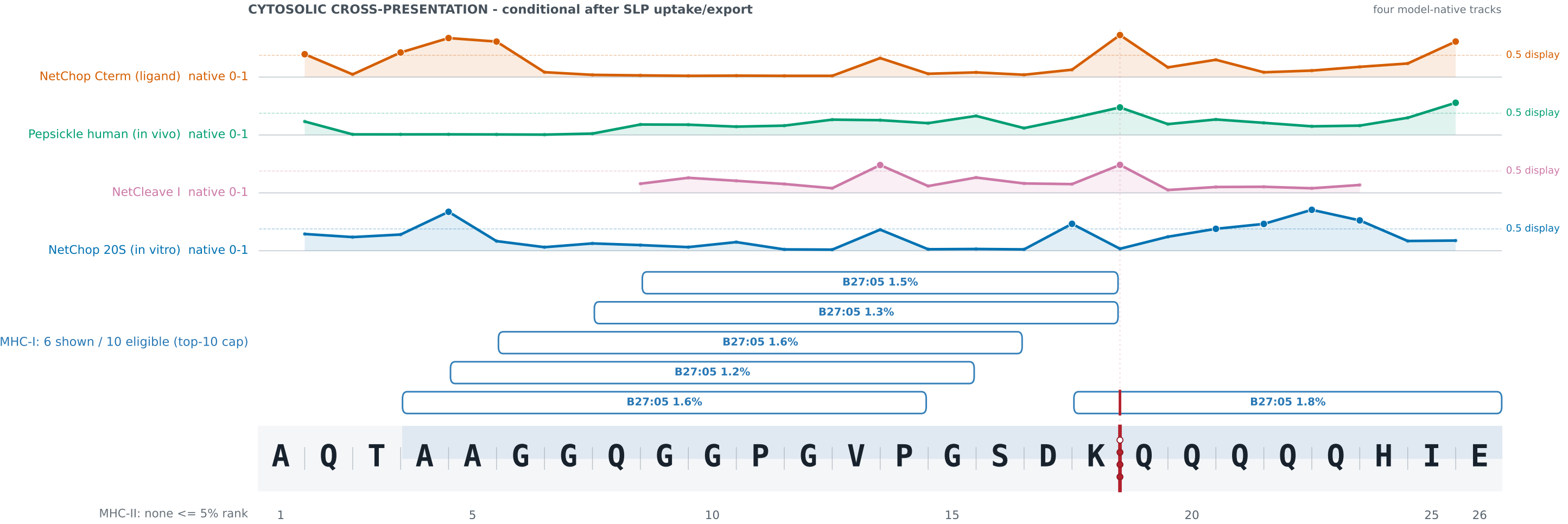
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

NR2F2 p.Gly65Val | JLF V2 | 26 aa

minimal epitope/window not disclosed for this SLP

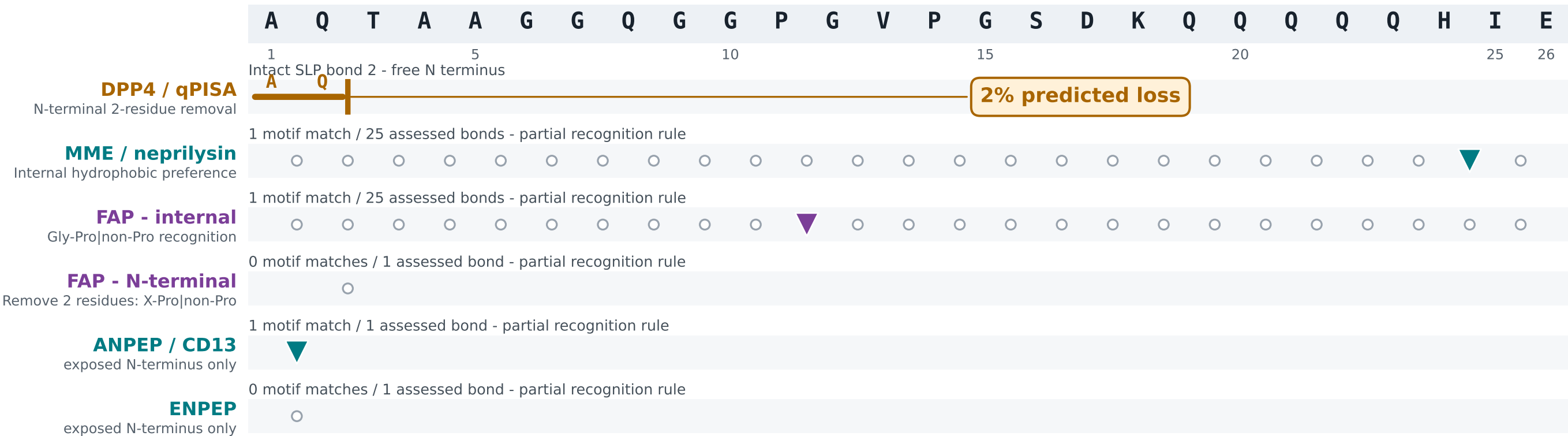


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

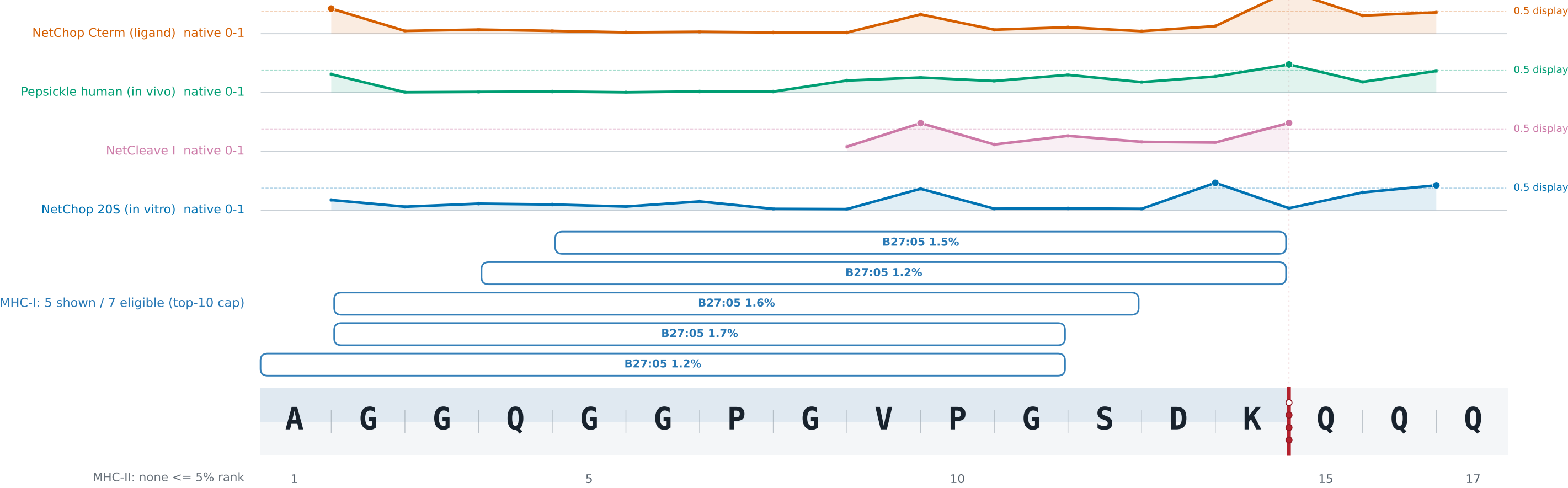
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

NR2F2 p.Gly65Val | CeGaT | 17 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

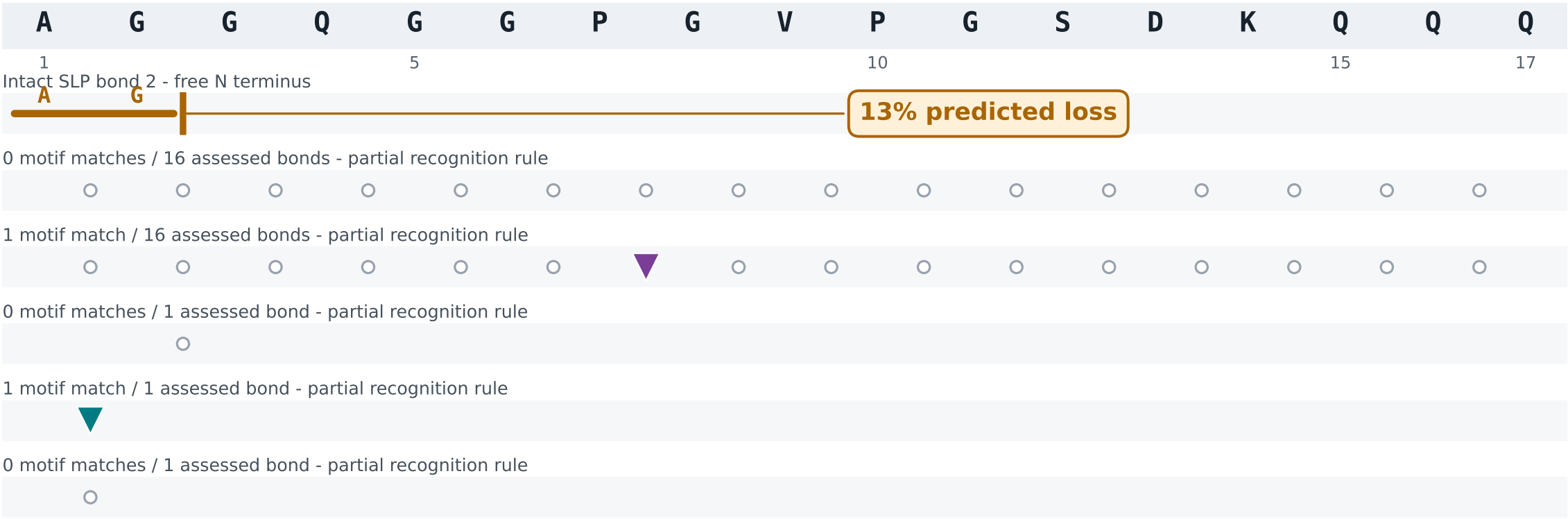


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

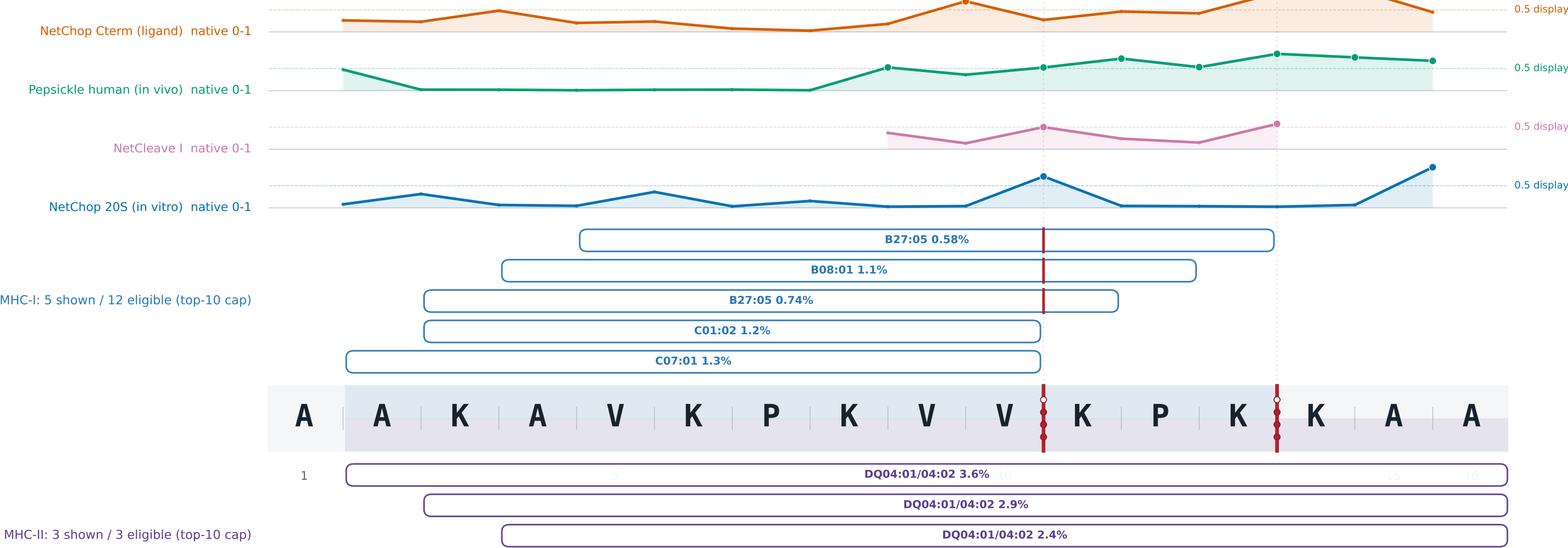
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

H1-2 p.Ala197_Lys201del | CeGaT | 16 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

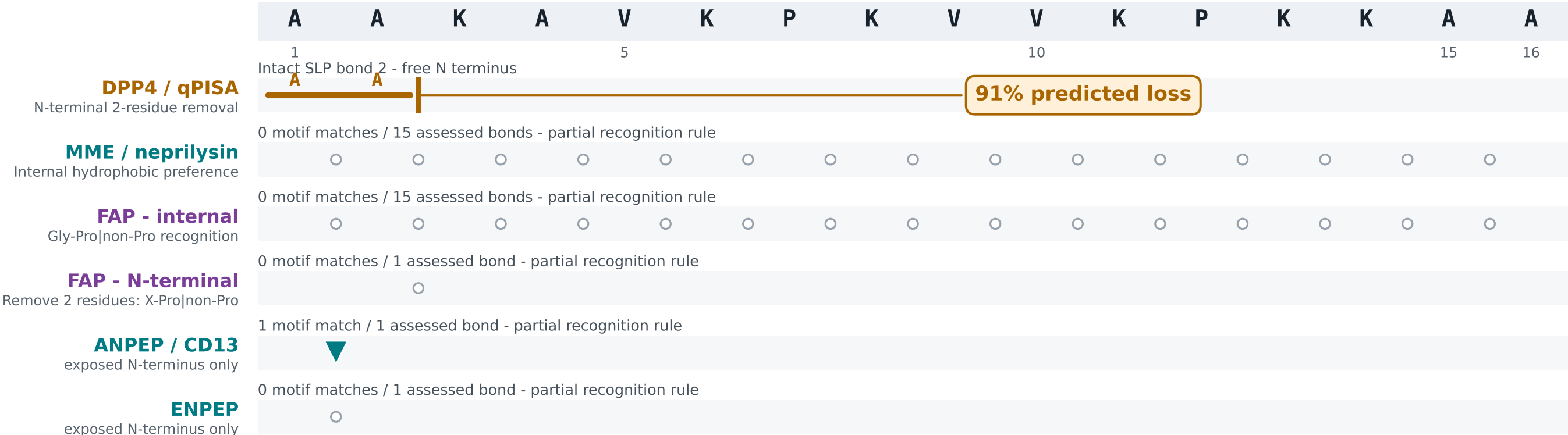


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

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